

Matter–Antimatter Asymmetry as the Unique Admissible Reconstruction of Persistent Cosmological Identity

Exchangeability Non-Transportability and the Boundary-Fixed Surplus Reconstruction Class

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Abstract

Matter–antimatter asymmetry is ordinarily treated as a dynamical problem of baryogenesis: which microphysical mechanisms generate a persistent baryonic surplus from an initially symmetric or nearly symmetric early universe. This paper does not propose a new baryogenesis mechanism, alter Standard-Model dynamics, or derive the baryon-to-photon ratio. It proves a narrower structural result. Once a cosmological history is treated as a coherent construction of persistent material identity, the admissibility kernel is instantiated automatically: reference, standing, admissibility, and irreversibility become non-optional constraints on same-domain continuation.

Within a fixed cosmological reconstruction category, exact matter–antimatter exchangeability is shown to be non-transportable through irreversible continuation whenever the correlations required to recover the exchange involution are no longer admissibly available on the late material-reference scope. Hidden reindexing, post-hoc symmetry restoration, carrier laundering, recoverability repair, and scope transport collapse into bookkeeping, inadmissible selector import, or larger-domain change. The resulting universal obstruction is represented by a nonvanishing exchangeability-totalization obstruction object.

The conclusion is not that standard baryogenesis programs are replaced, but that they are structurally reinterpreted as possible dynamical realizations of a more basic reconstruction requirement: persistent cosmological identity admits, up to bookkeeping-equivalent redescription, a unique admissible reconstruction class only after exact global exchangeability has failed as a standing-preserving transport invariant. The result fixes the admissible surplus reconstruction class and anchor-witness transport burden, not the numerical baryon abundance, baryon-to-photon ratio, CP phase values, or mechanism-specific production history. Nonvanishing of the burden-restricted obstruction gives witness nonemptiness on the burden support; the scope-indexed lower bound requires burden-wide exchange non-transportability on the declared material-reference burden. In physical realization, the class may be carried through standing-preserving transport, particle persistence, admissible CP-odd mechanism structure, and mass/inertia deformation-response profiles.

1 Introduction

The universe is not globally matter–antimatter symmetric in any observationally useful late-time sense. Standard baryogenesis theory treats that asymmetry as the output of specific early-universe dynamics constrained by the Sakharov conditions: baryon-number violation, C and CP violation, and departure from thermal equilibrium [1, 2, 3, 4]. That framework remains indispensable for

mechanism-building. The present paper asks a prior question: *why must some persistent anchor-witness asymmetry be admissibly fixed if cosmic history is to support stable material reference after irreversible evolution?*

This paper addresses that prior question as a same-domain non-survival theorem. It does not derive microphysics from a lower-level substrate, select a baryogenesis mechanism, infer the observed baryon-to-photon ratio, modify Standard Model dynamics, or posit CPT violation. Instead, it identifies the constraint already instantiated by cosmological regimes containing stable coherent material reference. Once a history contains compositional persistence, repeatable material interaction, retained structural identity, or observer-independent material reuse, the admissibility–standing–reference–irreversibility kernel is already active on that domain. The theorem is then: on such a scope, exact matter–antimatter exchangeability cannot survive irreversible history.

The paper’s core claim is conditional only on the target being a domain of stable coherent material reference, and structural:

Once a fixed cosmological scope requires persistent material reference through irreversible evolution, exact global matter–antimatter exchangeability cannot serve as the late same-history identity symmetry. When exchange non-transportability is burden-wide over the declared support, the admissible surplus reconstruction class has witness rank bounded below by the required anchor burden.

The key obstruction is not annihilation itself. Annihilation is dynamically permissible. Nor is the obstruction the mere fact that matter and antimatter have opposite bookkeeping labels. The obstruction is the attempted transport of exact exchange symmetry through irreversible history. Once a history has irreversibly lost the correlations needed for lawful global relabeling, an exact exchange symmetry can no longer be transported as a standing-preserving same-history identity symmetry into the late-time material-reference quotient. The theorem is therefore about carrier-role continuation, not about preferring one verbal label over another: the question is whether the late anchor classes that carry material-reference roles remain exchangeable without changing the same admissible lineage. That is where the contradiction appears.

The result is not an anthropic selection claim. It does not say that observers make the asymmetry likely, nor that observer existence explains a numerical value. It says that a declared late-time material-reference burden cannot be realized by a history whose exact matter–antimatter exchangeability remains globally transportable after irreversible loss. Observer-bearing lineages are one sufficient way to force the burden; chemistry-bearing material organization already forces a nonempty burden.

The theorem is not the tautology that a universe with persistent matter contains persistent matter. The nontrivial claim is that, once a late material-reference lineage is required to remain the same lineage after retrieval-irreversibility, exact global matter–antimatter exchangeability cannot remain a standing-preserving transport invariant unless an exchangeability preservation certificate remains available on the declared scope. If that certificate is absent, late material reference must be carried by a persistent anchor witness rather than by exact exchange symmetry; on sign-coherent material-standing scopes the sign of that witness is fixed along the same admissible lineage.

The paper operates at the reconstruction-governance level. It does not say that later microphysical processes are unreal, redundant, or unable to generate projected particle-number asymmetries in a physical model. It says that, once persistent material reference is the fixed target, those processes must occupy a reconstruction class whose admissibility is already forced by exchangeability non-

transportability. Later dynamics may realize, transfer, protect, dilute, or display projected particle-number asymmetry; it does not generate the admissibility of the admissible surplus reconstruction class from a previously exact same-scope exchangeable history.

The claim status is therefore fixed. The paper proves existence, sign fixation, lower-bound burden, quotient uniqueness, and non-transportability of exact exchangeability in the declared material-reference scope. It does not compute η_B , derive a baryon-to-photon ratio, determine CP numerical values, privilege a quark sector, or select leptogenesis, electroweak baryogenesis, freeze-out, reheating, or any other physical mechanism.

To make this precise, the paper introduces an explicit anchor layer: persistent anchor candidates, downstream material-reference role profiles, anchor equivalence, a canonical anchor quotient, anchor witness sets, and anchor rank. The result is not merely “it has to be that way.” Rather, it is:

exact matter–antimatter symmetry cannot support persistent material identity once irreversible evolution destroys the admissible transportability of the exchange symmetry through late-time anchor structure.

Three structural consequences follow. First, exact symmetric persistence fails. Second, a nonzero admissible surplus reconstruction class, represented by a persistent anchor witness, is obtained on the burden support, and it satisfies a scope-indexed lower bound under burden-wide exchange non-transportability. Third, once that witness becomes standing-bearing on a sign-coherent material-standing scope, its sign class is fixed along the same admissible lineage and cannot later re-symmetrize without illicit reindexing or scope change.

irreversibility $\implies$ no lawful global relabeling $\implies$ anchor fixation $\implies$ persistent asymmetry

Figure 1: Proof spine in one line. The theorem is carried not by annihilation alone, but by the failure of exact exchange symmetry to remain lawfully transportable through irreversible history into the late-time anchor quotient.

2 Exact Load, Kernel Instantiation, and Fixed-Domain Scope

2.1 Method posture

The paper is structural. It introduces no new particle content, no modified dynamics, no external selector principle, and no probabilistic typicality postulate. It works only with:

1. a fixed cosmological history type,
2. lawful identity-preserving redescription,
3. irreversible loss of retrievable correlation,
4. standing conservation along admissible lineage, and
5. the requirement that persistent material reference remain standing-bearing on the declared scope.

The theorem is therefore conditional only in the ordinary mathematical sense: it applies to domains that instantiate stable coherent material reference. The burden is not a discretionary semantic preference once the target domain already contains persistent compositional organization, repeatable material interaction, or retained material identity. In such a domain, the kernel is already instantiated and the admissibility constraints in this theorem package are forced.

Remark 2.1 (Local AASC vocabulary). The only AASC vocabulary needed in the paper is used in its fixed-domain sense. *Admissibility* distinguishes licensed from unlicensed same-scope continuation. *Standing* is reusable status under later use. *Reference* fixes which material carrier class is being continued. *Irreversibility* blocks same-history post-hoc repair after the relevant correlation structure has been lost. The paper uses no additional enforcement tool.

Definition 2.2 (Convention: meaning of nonzero). Throughout, “nonzero” means a nonempty standing-bearing witness class in the anchor quotient, or a nonvanishing obstruction in the declared reconstruction category, unless numerical particle abundance is explicitly named. It does not mean a derived baryon-to-photon ratio, baryon density, or mechanism-level particle-number value.

2.2 Automatic kernel instantiation in persistent physical domains

Definition 2.3 (Stable coherent material-reference domain). A cosmological domain is a *stable coherent material-reference domain* when its physical use requires at least one of the following to be preserved under lawful continuation:

1. repeatable material interaction involving identifiable carrier classes,
2. compositional persistence of bound or organized structures,
3. retained structural identity across later reactions, records, or material reuse,
4. observer-independent coherent reference to the same material organization across a history segment.

The definition is physical rather than semantic: it concerns whether later interactions, compositions, or records are treated as involving the same material carrier lineage.

Theorem 2.4 (Kernel instantiation theorem for persistent cosmological structure). *Let a cosmological domain instantiate stable coherent material reference. Then the admissibility–standing–reference–irreversibility kernel is automatically instantiated on that domain. In particular:*

1. *admissibility is required to distinguish licensed from unlicensed continuations of the same carrier lineage;*
2. *standing is required for subsequent uses of that lineage to count as uses of the same material content;*
3. *reference is required to determine which carrier class is being continued;*
4. *irreversibility is required to prevent failed or lost identity transport from being retroactively repaired as the same history.*

Thus the kernel is not an additional interpretive overlay on cosmology. It is the necessity constraint already instantiated by any domain requiring stable coherent material reference.

Proof. Stable coherent material reference requires later states, interactions, or records to be about the same material carrier lineage rather than merely about a replacement or relabeling. If licensed and unlicensed continuations cannot be distinguished, then the domain has no admissible criterion for preserving the same lineage; admissibility is required. If a carrier lineage cannot retain reusable status under later use, then no later reaction, record, or composite structure can count as involving the same material content; standing is required. If there is no determinate carrier class being continued, then lawful continuation has no fixed object; reference is required. Finally, if an identity-transport failure can be reauthored after correlation loss as the same successful history, then success and failure were never determinate on the fixed domain; irreversibility is required. These four conditions are therefore forced by the target domain itself. $\square$

Corollary 2.5 (Persistent carrier realization). *In a stable coherent material-reference domain, persistent carrier structure is physically load-bearing rather than a matter of naming convention. Any distinction that supports repeatable material interaction, compositional persistence, or retained structural identity is anchor or tensor content on that domain; if it supports none of these, it is skin.*

Proof. By the kernel instantiation theorem, later use of the same carrier lineage requires admissibility, standing, reference, and irreversibility. A distinction that supports such later use contributes to the admissible identity of the carrier lineage and is therefore anchor or tensor content. A distinction that contributes nothing to admissible continuation, standing, reference, or later interaction changes no standing-bearing content and is skin. $\square$

Corollary 2.6 (Exchange-transport alternative from kernel instantiation). *If irreversible cosmological evolution and stable coherent material reference coexist on one fixed domain, then exact global matter–antimatter exchangeability must either remain lawfully transportable through the irreversible stage or fail as a same-domain identity symmetry. When irreversible loss blocks such transport, the exact-exchange route fails. The later obstruction and realization theorem package proves witness nonemptiness on the burden support, and proves the scope-indexed lower bound only under burden-wide exchange non-transportability; sign fixation is obtained only on SCMS scopes.*

Proof. Stable coherent material reference instantiates the kernel. Irreversible cosmological evolution blocks same-history post-hoc repair after the correlations required for global exchange transport are lost. Exact exchangeability can therefore survive only if the exchange map itself remains lawfully transportable as a reference-preserving identity symmetry. If the map cannot be transported, exact exchange no longer supplies the same-domain identity symmetry. The burden-restricted obstruction and anchor-realization results establish witness nonemptiness in the subsequent obstruction and realization theorems; the burden-wide form supplies the lower-bound conclusion. This corollary records only the initial transport alternative forced by the kernel. $\square$

Remark 2.7 (Effect on the scope-burden language). The later phrase “scope burden” should not be read as a discretionary preference. It denotes the burden already induced when the target domain contains stable coherent material reference. A domain may decline that target, but it then no longer supplies a same-scope counterexample to a theorem about persistent material organization.

2.3 Boundary-regime dependency orientation

At the dependency level, the admissible surplus reconstruction class belongs to the first licensed reconstruction package in which persistent material reference, retrieval status, continuation, anchor quotient, role profile, and surplus witness are jointly available. The formal boundary object is introduced after the anchor quotient and reconstruction category have been defined, so that the boundary class is specified only after its data exist. The boundary is a licensing status in the reconstruction order, not a first temporal instant, causal event, thermodynamic epoch, or baryogenesis mechanism.

2.4 Fixed cosmological scope

Definition 2.8 (Fixed cosmological scope). A *fixed cosmological scope* $\mathcal{S}$ consists of:

1. a class $\mathcal{H}$ of candidate cosmic histories,
2. a lawful-redescription relation preserving the same history as a history of the same material structures,
3. a designated family $\mathcal{R}$ of standing-bearing material reference classes that must remain well-defined along those histories, and
4. an irreversibility condition forbidding post-hoc recovery of a history's reference structure by global relabeling once the necessary retrievable correlations have been lost.

Definition 2.9 (Matter and antimatter carrier classes). For a history $h \in \mathcal{H}$, write $M_t^M(h)$ and $M_t^{\bar{M}}(h)$ for the matter-class and antimatter-class carriers present at stage t of the history. These are bookkeeping classes, not additional ontic primitives: their role is to track standing-bearing material identity across the declared scope.

Definition 2.10 (Persistent material reference). A reference class $r \in \mathcal{R}$ is *persistently material* on scope $\mathcal{S}$ if there exists an admissible lineage through which r remains attached to surviving material structures under lawful redescription, without requiring a later global relabeling of matter and antimatter labels.

Definition 2.11 (Declared material-reference burden). The declared material-reference burden of a fixed scope $\mathcal{S}$ is the subfamily

$$\mathcal{R}_{\text{bur}} \subseteq \mathcal{R}$$

of standing-bearing material-reference classes that the history must realize after retrieval-irreversibility. The required anchor burden $\rho_{\mathcal{S}}$ is the minimum number of distinct persistent anchor classes required to realize all classes in $\mathcal{R}_{\text{bur}}$ on the post-irreversible scope.

Definition 2.12 (Sign-coherent material-standing scope). A *sign-coherent material-standing scope*, abbreviated *SCMS scope*, is a stable coherent material-reference scope whose late material-reference burden is connected under one sign-side anchor organization. Equivalently, the standing-bearing material-reference classes in $\mathcal{R}$ participate in one connected late material organization under lawful continuation, without cross-sign annihilative reset or global reindexing.

Remark 2.13 (Status of SCMS connectivity). SCMS connectivity is not a baryogenesis mechanism and not a preference for the sign later named “matter.” It records the connectedness burden imposed by a chemistry-bearing or observer-bearing material organization. Disconnected matter-rich and antimatter-rich regions can be treated by indexed scopes unless a single SCMS burden is explicitly declared.

Definition 2.14 (Chemistry-bearing and observer-bearing scope burden). A scope $\mathcal{S}$ carries *chemistry-bearing persistent organization* if it admits at least one late-time lineage in which material carriers must remain reidentifiable through composition, retention, or reaction. It carries *observer-bearing late-time history* if it admits at least one lineage in which later record-bearing material structures must remain reidentifiable as belonging to the same continuing material organization.

Theorem 2.15 (Scope-burden bridge theorem). *If a fixed cosmological scope $\mathcal{S}$ carries chemistry-bearing persistent organization or observer-bearing late-time history, then:*

1. $\mathcal{R} \neq \emptyset$, and
2. the required anchor burden satisfies $\rho_{\mathcal{S}} \geq 1$.

Equivalently, such a scope necessarily demands at least one reidentifiable material carrier class after irreversible evolution.

Proof. Chemistry-bearing persistent organization requires at least one material carrier class to remain reidentifiable through later composition or reaction; otherwise there is no determinate material organization that persists through the history. Likewise, observer-bearing late-time history requires at least one continuing material carrier class to support retained record-bearing organization; otherwise there is no determinate material lineage to which later record states belong. In either case there is at least one standing-bearing material reference class, so $\mathcal{R} \neq \emptyset$.

By definition, $\rho_{\mathcal{S}}$ is the minimum number of distinct persistent anchor classes required to realize the declared burden $\mathcal{R}_{\text{bur}}$ after irreversible evolution. Since a chemistry-bearing or observer-bearing scope admits at least one burden-bearing material-reference class, $\mathcal{R}_{\text{bur}} \neq \emptyset$ and $\rho_{\mathcal{S}} \geq 1$. $\square$

Remark 2.16 (Why the burden is not optional). The lower-bound theorem is not a free scope preference. It is a structural lower bound relative to the declared burden $\mathcal{R}_{\text{bur}}$ that is already forced once the scope includes chemistry-bearing or observer-bearing persistent organization. A history may avoid the theorem only by refusing that burden; it then ceases to be a same-scope cosmological history of persistent material organization.

2.5 Retrieval-irreversibility is not microscopic time-asymmetry

Definition 2.17 (Irreversible loss of retrievable correlation). A history h exhibits *irreversible loss of retrievable correlation* on $\mathcal{S}$ if there exists a stage $t_*(h)$ such that for all later stages $t \geq t_*(h)$ the data required to reconstruct a global matter–antimatter relabeling as a lawful same-history symmetry is no longer recoverable from the history on the fixed scope.

Definition 2.18 (Dynamical reversibility versus retrieval-irreversibility). A history is *dynamically reversible* when its microscopic equations admit a formal inverse evolution on an exact state specification. A history is *retrieval-irreversible on $\mathcal{S}$* when the fixed scope no longer contains the correlations required to transport a global relabeling as a lawful identity-preserving map. The present theorem requires only retrieval-irreversibility.

Proposition 2.19 (Loschmidt and recurrence insulation). *The argument does not depend on fundamental microscopic irreversibility, violation of unitary reversibility, or denial of Poincare-type recurrence in idealized systems. It requires only that the declared cosmological scope fail to contain a lawful recovery map for the global matter–antimatter reindexing after the relevant correlations have been dispersed.*

Proof. A microscopic inverse evolution, if supplied by an exact external state description, is not automatically a lawful same-scope recovery of standing-bearing material-reference structure. The latter requires the late history itself, under its declared admissible data and redescription rules, to carry the information needed to reconstruct the global matter–antimatter exchange as the same-history identity map. If that information is absent on the fixed scope, an external reconstruction would import hidden data or change the scope. Therefore microscopic reversibility does not restore admissible global reindexing. $\square$

Proposition 2.20 (Structural rather than anthropic conditioning). *The use of chemistry-bearing or observer-bearing late-time history is a scope-burden condition, not an anthropic probability weighting. The theorem classifies which histories can support persistent material reference; it does not infer probabilities from observer existence.*

Proof. An anthropic argument weights or selects histories by the fact that observers occur in them. The present argument instead fixes a target burden: persistent material reference under irreversible continuation. Chemistry-bearing structure or observer-bearing history are sufficient conditions for that burden. Once the burden is fixed, the proof uses only lawful redescription, irreversible loss, and anchor realization. No probability measure over observers or histories enters the theorem. $\square$

3 Anchor Formalization and Canonical Anchor Quotient

Definition 3.1 (Persistent anchor candidate). Fix a history $h \in \mathcal{H}$ and a stage $t \geq t_*(h)$ after irreversible loss has occurred. A *persistent anchor candidate* is a surviving matter-side or antimatter-side carrier configuration in $M_t^M(h) \sqcup M_t^{\bar{M}}(h)$ that is eligible, in principle, to support a standing-bearing material reference class on the fixed scope.

Definition 3.2 (Downstream material-reference role map). Let $\mathcal{C}_S(h)$ be the set of persistent anchor candidates of h after the irreversible stage. For $a \in \mathcal{C}_S(h)$, define the *downstream material-reference role map*

$$\text{Role}_h^S(a) \subseteq \mathcal{R}$$

to be the set of standing-bearing material reference classes that a can lawfully realize under late-time continuation on the fixed scope.

Definition 3.3 (Anchor equivalence). Let $a, b \in \mathcal{C}_S(h)$. We say that a and b are *anchor-equivalent*, written

$$a \sim_{S,h}^{\text{anch}} b,$$

if:

1. a and b have the same sign type (both matter-side or both antimatter-side), and
2. $\text{Role}_h^S(a) = \text{Role}_h^S(b)$.

Thus anchor equivalence is fixed by the full downstream material-reference role profile together with sign type.

Definition 3.4 (Persistent anchor class). A *persistent anchor class* is an equivalence class of persistent anchor candidates under $\sim_{\mathcal{S},h}^{\text{anch}}$. Define the total anchor quotient

$$\mathcal{Q}_{\mathcal{S}}(h) := \mathcal{C}_{\mathcal{S}}(h) / \sim_{\mathcal{S},h}^{\text{anch}}.$$

Write

$$\mathcal{A}_{\mathcal{S}}^{\text{M}}(h), \quad \mathcal{A}_{\mathcal{S}}^{\text{M}}(h)$$

for the matter-side and antimatter-side subquotients of $\mathcal{Q}_{\mathcal{S}}(h)$.

Definition 3.5 (Burden-realizing anchor support). The burden-realizing post-irreversible anchor support is the subquotient

$$\mathcal{Q}_{\mathcal{S}}^{\text{bur}}(h) := \{[a] \in \mathcal{Q}_{\mathcal{S}}(h) : \text{Role}_h^{\mathcal{S}}(a) \cap \mathcal{R}_{\text{bur}} \neq \emptyset\}.$$

It contains exactly those persistent anchor classes that realize at least one class in the declared material-reference burden. Exchangeability failure outside $\mathcal{Q}_{\mathcal{S}}^{\text{bur}}(h)$ is not sufficient for the lower-bound result; the relevant obstruction must strike the burden-realizing support.

Definition 3.6 (Burden-realizing subfamilies). A subfamily $U \subseteq \mathcal{Q}_{\mathcal{S}}^{\text{bur}}(h)$ is *burden-realizing* when the role profiles of its members cover the declared material-reference burden:

$$\mathcal{R}_{\text{bur}} \subseteq \bigcup_{[a] \in U} \text{Role}_h^{\mathcal{S}}(a).$$

It is *inclusion-minimal* when no proper subfamily of U is burden-realizing. It is *minimum* when its cardinality is $\rho_{\mathcal{S}}$, the least cardinality of any burden-realizing subfamily. By definition, every burden-realizing subfamily has cardinality at least $\rho_{\mathcal{S}}$, and when the declared burden is realized there is at least one minimum burden-realizing subfamily of cardinality $\rho_{\mathcal{S}}$.

Definition 3.7 (Burden-wide genuine retrieval-irreversibility). The declared burden $\mathcal{R}_{\text{bur}}$ is *burden-wide genuinely retrieval-irreversible* on $\mathcal{Q}_{\mathcal{S}}^{\text{bur}}(h)$ when every member of every minimum burden-realizing subfamily $U \subseteq \mathcal{Q}_{\mathcal{S}}^{\text{bur}}(h)$ lies outside the admissible domain of exact post-irreversible exchange transport preserving its role profile into $\mathcal{R}_{\text{bur}}$. Equivalently, every anchor class in a minimum-cardinality realization of the declared burden lacks an exchangeability preservation certificate on the declared burden support.

Definition 3.8 (Anchor presentation). An *anchor presentation* for $(\mathcal{S}, h)$ is a triple (X, r, Role_X) where:

1. X is a set intended to serve as the anchor state space,
2. $r : \mathcal{C}_{\mathcal{S}}(h) \rightarrow X$ is a surjection, and
3. for every $q \in \mathcal{R}$ there exists a predicate P_q^X on X such that

$$q \in \text{Role}_h^{\mathcal{S}}(a) \iff P_q^X(r(a)) \quad (a \in \mathcal{C}_{\mathcal{S}}(h)).$$

The presentation is *admissible* if it is sound and complete with respect to anchor equivalence:

$$a \sim_{\mathcal{S},h}^{\text{anch}} b \iff r(a) = r(b).$$

Lemma 3.9 (Anchor quotient universal property). *Let $q : \mathcal{C}_{\mathcal{S}}(h) \rightarrow \mathcal{Q}_{\mathcal{S}}(h)$ be the quotient map. If F is any function on anchor candidates that is constant on anchor-equivalence classes, then there exists a unique induced map*

$$\tilde{F} : \mathcal{Q}_{\mathcal{S}}(h) \rightarrow Y$$

such that $F = \tilde{F} \circ q$.

Proof. Define $\tilde{F}([a]) := F(a)$. This is well-defined because F is constant on anchor-equivalence classes. Uniqueness follows from surjectivity of q . $\square$

Theorem 3.10 (Anchor-quotient uniqueness theorem). *Let (X, r, Role_X) be any admissible anchor presentation for $(\mathcal{S}, h)$. Then there exists a unique bijection*

$$\phi : \mathcal{C}_{\mathcal{S}}(h) / \sim_{\mathcal{S},h}^{\text{anch}} \longrightarrow X, \quad \phi([a]) = r(a).$$

In particular, the anchor quotient is canonical up to unique isomorphism: any admissible same-scope presentation of persistent anchor content is uniquely equivalent to the quotient by anchor equivalence.

Proof. Because the presentation is admissible, $r(a) = r(b)$ iff $a \sim_{\mathcal{S},h}^{\text{anch}} b$. Hence $\phi([a]) = r(a)$ is well-defined, injective, and surjective. Uniqueness is exactly the quotient universal property. $\square$

Definition 3.11 (Candidate anchor-witness notation and ranks). Define the matter-side and antimatter-side *candidate anchor-witness sets*

$$\mathcal{W}_{\mathcal{S}}^{\text{M}}(h) \subseteq \mathcal{A}_{\mathcal{S}}^{\text{M}}(h), \quad \mathcal{W}_{\mathcal{S}}^{\bar{\text{M}}}(h) \subseteq \mathcal{A}_{\mathcal{S}}^{\bar{\text{M}}}(h)$$

to consist of those persistent anchor classes whose exchanged counterpart is *not* post-irreversibly admissibly transportable through the history while preserving the same downstream material-reference role profile. These sets may be empty at the level of notation. Their nonemptiness is extracted only after the burden-restricted exchange-totalization obstruction is combined with anchor realization and the declared material-reference burden.

The total candidate anchor-witness family is

$$\mathcal{W}_{\mathcal{S}}(h) := \mathcal{W}_{\mathcal{S}}^{\text{M}}(h) \sqcup \mathcal{W}_{\mathcal{S}}^{\bar{\text{M}}}(h).$$

The burden-restricted candidate anchor-witness family is

$$\mathcal{W}_{\mathcal{S}}^{\text{bur}}(h) := \mathcal{W}_{\mathcal{S}}(h) \cap \mathcal{Q}_{\mathcal{S}}^{\text{bur}}(h).$$

The *candidate anchor rank* and *burden-restricted anchor rank* are

$$\text{arank}_{\mathcal{S}}(h) := |\mathcal{W}_{\mathcal{S}}^{\text{M}}(h)| + |\mathcal{W}_{\mathcal{S}}^{\bar{\text{M}}}(h)|, \quad \text{arank}_{\mathcal{S}}^{\text{bur}}(h) := |\mathcal{W}_{\mathcal{S}}^{\text{bur}}(h)|.$$

The *signed witness coordinate* is

$$\Delta_B^{\mathcal{S}}(h) := |\mathcal{W}_{\mathcal{S}}^{\text{M}}(h)| - |\mathcal{W}_{\mathcal{S}}^{\bar{\text{M}}}(h)|.$$

Here $\Delta_B^S(h)$ is a signed coordinate of the witness notation rather than a sign-fixation theorem. Section 11 establishes sign fixation for SCMS scopes. The required anchor burden ρ_S is defined by the declared burden $\mathcal{R}_{\text{bur}}$.

Remark 3.12 (Counting convention). The paper counts a canonical quotient object. It does not count particles, arbitrary sectors, or vague “lineages.” It counts persistent anchor classes modulo the full downstream material-reference role profile and sign type. Numerical particle abundances belong to downstream dynamical modeling, not to this structural reconstruction theorem.

Reconstruction transition. The anchor quotient fixes the object counted by the theorem. The reconstruction category then places that quotient inside a fixed comparison class, so that the asymmetry claim is stated as a universal obstruction and reconstruction theorem over a locked comparison class rather than as an informal structural observation.

4 The Pre-Witness Reconstruction Category of Admissible Persistent-Carrier Histories

This section fixes the mathematical comparison class over which the obstruction claim is first proved. The category is intentionally scope-locked. It formalizes the persistent-carrier reconstruction problem rather than a full dynamical model of baryogenesis.

The comparison class is not chosen to force the conclusion. The first category is a *pre-witness* reconstruction category: it preserves the kernel, anchor quotient, role profiles, continuation, retrieval status, skin covariance, and no-selector/no-repair discipline, but it does not impose preservation of the signed witness coordinate as an equivalence condition. Witness preservation enters only in the audit category, after obstruction-to-witness extraction has registered the witness predicate and established nonempty burden-restricted witness support where $\mathcal{O}_{\text{ex}}^{\text{bur}} \neq 0$; burden-wide extraction supplies the lower-bound coordinate where its stronger hypothesis is assumed.

Definition 4.1 (Pre-witness baryogenesis reconstruction category). Let $\mathcal{C}_{\text{BAR}}^{\text{pre}}$ be the category whose objects are finite persistent-carrier reconstruction systems

$$B^{\text{pre}} = (\mathcal{S}, h, t_*, \mathcal{C}_S(h), \mathcal{Q}_S(h), \Xi_h, \text{Cont}_h, \text{Rec}_\Xi(h), \text{Env}_S).$$

Here $\mathcal{S}$ is a fixed cosmological material-reference scope; $h \in \mathcal{H}$ is a candidate history; $t_* = t_*(h)$ is the declared exchange-relevant retrieval-irreversibility stage; $\mathcal{C}_S(h)$ is the post-irreversible persistent anchor-candidate set; $\mathcal{Q}_S(h)$ is the anchor quotient; Ξ_h is the pre-irreversible matter–antimatter exchange involution on the exchangeable prefix when such a prefix exists; Cont_h is the admissible late-time continuation relation; $\text{Rec}_\Xi(h)$ records whether an exchangeability preservation certificate is available from the admitted late data; and Env_S records envelope data not derived by the theorem, including dynamical mechanism data, thermal history, interaction rates, and cosmological approximation class. The persistent witness family $\mathcal{W}_S(h)$ and signed witness $\Delta_B^S(h)$ are not protected morphism data in $\mathcal{C}_{\text{BAR}}^{\text{pre}}$.

Definition 4.2 (Pre-witness admissible reconstruction morphism). A morphism $\Phi : B^{\text{pre}} \rightarrow B'^{\text{pre}}$ in $\mathcal{C}_{\text{BAR}}^{\text{pre}}$ is an admissible persistent-carrier reconstruction map satisfying the following conditions.

- M1. Kernel-anchor preservation.** Φ preserves the instantiated admissibility–standing–reference–irreversibility kernel and sends persistent anchor candidates to persistent anchor candidates without changing the fixed material-reference burden.
- M2. Anchor-quotient preservation.** $a \sim_{S,h}^{\text{anch}} b$ iff $\Phi(a) \sim_{S',h'}^{\text{anch}} \Phi(b)$, and the induced map on $\mathcal{Q}_S(h)$ preserves sign-sensitive anchor classes unless sign is explicitly relabeled as skin after quotient preservation.
- M3. Role-profile preservation.** For every anchor class $[a]$, the downstream material-reference role profile is preserved:
- $$\text{Role}_h^S(a) = \text{Role}_{h'}^{S'}(\Phi(a))$$
- up to lawful relabeling of reference names that leaves the material-reference roles fixed.
- M4. Continuation preservation.** If $[a] \text{Cont}_h [b]$ is an admissible late-time continuation relation, then $[\Phi(a)] \text{Cont}_{h'} [\Phi(b)]$ is admissible, and conversely for morphisms claimed as equivalences.
- M5. Retrieval-status preservation.** Φ preserves whether the pre-irreversible exchange involution Ξ_h admits a post-irreversible exchangeability preservation certificate. A morphism may not turn a nonrecoverable exchange map into a recoverable one without changing scope.
- M6. Skin covariance.** Coordinate descriptions, naming conventions, representative choices, and equivalent history charts may vary only when anchor quotient, role profile, continuation relation, and retrieval status are preserved.
- M7. No-selector and no-repair discipline.** Φ may not introduce a hidden global index, canonical representative, typicality measure, preferred sign side, or late post-hoc relabeling that fixes standing-bearing material reference outside the declared anchor quotient.

4.1 Why the pre-witness morphisms are not definitionally loaded

The morphism conditions in $\mathbb{C}_{\text{BAR}}^{\text{pre}}$ are not added to force the conclusion. Each condition protects one component of the fixed persistent-material reconstruction target. Deleting any condition either changes the theorem object, permits hidden selector import, permits post-hoc repair, or collapses material-reference identity into presentation skin.

Morphism condition	If removed, the comparison fails because
M1 Kernel-anchor preservation	the comparison no longer concerns the same admissible material-reference construction.
M2 Anchor-quotient preservation	persistent carrier identity can be changed while pretending the target is fixed.
M3 Role-profile preservation	“same material carrier” becomes underdetermined.
M4 Continuation preservation	lineage can change silently.
M5 Retrieval-status preservation	retrieval-irreversibility can be repaired by definition.
M6 Skin covariance	notation, chart, or name changes can masquerade as tensor changes.
M7 No-selector/no-repair discipline	hidden global indices and post-hoc exchange restoration re-enter.

Proposition 4.3 (Minimality of pre-witness morphism discipline). *Each $\mathbb{C}_{\text{BAR}}^{\text{pre}}$ morphism condition protects one component of the fixed persistent-material reconstruction target. Removing any one*

condition either changes the theorem object, permits hidden selector import, permits post-hoc repair, or collapses material-reference identity into presentation skin.

Proof. A same-scope reconstruction comparison must preserve the admissibility kernel, the late anchor quotient, the role profiles that identify material-reference use, the continuation relation, the retrieval status, and the distinction between tensor content and skin. If any one of these is not protected, an alleged morphism can either change the target while retaining the same name, repair a failed exchange operation, add a hidden selector, or erase carrier identity by convention. Therefore the listed conditions are the minimal pre-witness comparison-class discipline for the declared target. $\square$

Proposition 4.4 ($\mathcal{C}_{\text{BAR}}^{\text{pre}}$ is a category). *Finite pre-witness baryogenesis reconstruction objects together with admissible pre-witness reconstruction morphisms form a category.*

Proof. The identity morphism preserves every component of B^{pre} : the kernel, anchor candidates, quotient classes, role profiles, continuation relation, retrieval status, envelope ledger, and no-selector/no-repair discipline. For composition, let $\Phi : B^{\text{pre}} \rightarrow B'^{\text{pre}}$ and $\Psi : B'^{\text{pre}} \rightarrow B''^{\text{pre}}$ be admissible. Preservation of anchor equivalence, role profiles, continuation, and retrieval status is transitive. A hidden selector, repair, or scope transport cannot appear in $\Psi \circ \Phi$ unless it appears in one of the two factors, which admissibility forbids. Associativity is inherited from ordinary composition of the underlying maps. Thus $\mathcal{C}_{\text{BAR}}^{\text{pre}}$ is a category. $\square$

Definition 4.5 (Pre-witness admissible quotient). The pre-witness admissible quotient category $\mathcal{Q}_{\text{BAR}}^{\text{pre}}$ has objects the equivalence classes of $\mathcal{C}_{\text{BAR}}^{\text{pre}}$ under morphisms preserving kernel instantiation, anchor quotient, role profile, continuation, retrieval status, skin covariance, and no-selector/no-repair discipline. It does not quotient by preservation of $\mathcal{W}_S(h)$ or $\Delta_B^S(h)$.

Proposition 4.6 (Witness preservation is not an input to witness existence). *The nonvanishing exchange-totalization obstruction is proved in $\mathcal{C}_{\text{BAR}}^{\text{pre}}$, before witness-preserving equivalence is imposed. Obstruction-to-witness extraction then registers the witness predicate and establishes nonempty burden-restricted witness support where $\mathcal{O}_{\text{ex}}^{\text{bur}} \neq 0$ on a nonempty burden support; the scope-indexed lower-bound coordinate requires the separate burden-wide hypothesis. The witness-preserving category $\mathcal{C}_{\text{BAR}}^{\text{wit}}$ is an audit quotient for comparing reconstructions after the relevant witness data have been registered; it is not a premise of the proof that the witness predicate applies.*

Proof. The pre-witness category preserves exactly the data needed to pose the exchange-transport problem: anchor quotient, role profiles, continuation, and retrieval status. The exchange-totalization obstruction is then evaluated on that data. Only after obstruction-to-witness extraction registers the witness predicate and supplies nonempty burden-restricted witness support does the audit category require preservation of witness data. The signed coordinate is retained as audit data; its sign-fixation status is established only on SCMS scopes. Therefore the proof of witness applicability does not assume the witness as morphism-preserved output. $\square$

5 Exchange Transport as a Partial Operator and Obstruction Object

The pre-witness category controls admissible reconstruction at the object level. The exchange layer makes the exchangeability obstruction intrinsic by treating exact matter–antimatter exchange as

a partial operator whose domain is fixed before retrieval-irreversibility. A post-irreversible exact symmetry claim is then an attempted totalization of that partial operator on the declared carrier.

Definition 5.1 (Partial exchange-transport object). Let $\mathcal{C}_{\text{BAR}}^\partial$ be the partial reconstruction category whose objects are tuples

$$B^\partial = (B, X_h, \mathsf{X}_h, E_{\text{pre}}, E_{\text{post}}),$$

where $B \in \text{Obj}(\mathcal{C}_{\text{BAR}}^{\text{pre}})$, X_h is the material-reference carrier of the history, E_{pre} is the pre-irreversible exchangeable prefix envelope, E_{post} is the post-irreversible anchor envelope, and

$$\mathsf{X}_h : X_h \rightharpoonup X_h$$

is the partial matter–antimatter exchange operator with

$$\text{dom}(\mathsf{X}_h) = E_{\text{pre}} \cup E_{\text{post}}^{\text{cert}}.$$

Here $E_{\text{post}}^{\text{cert}} \subseteq E_{\text{post}}$ is nonempty only when an exchangeability preservation certificate exists. Under genuine retrieval-irreversibility, $E_{\text{post}}^{\text{cert}} = \emptyset$ for the standing-bearing anchor quotient.

Definition 5.2 (Domain-preserving partial reconstruction morphism). A morphism $\Phi^\partial : B^\partial \rightarrow B'^\partial$ in $\mathcal{C}_{\text{BAR}}^\partial$ is a pre-witness admissible morphism $\Phi : B \rightarrow B'$ in $\mathcal{C}_{\text{BAR}}^{\text{pre}}$ that also preserves partial exchange-operator domains:

$$\Phi(E_{\text{pre}}) \subseteq E'_{\text{pre}}, \quad \Phi(E_{\text{post}}) \subseteq E'_{\text{post}}, \quad \Phi(\text{dom}(\mathsf{X}_h)) \subseteq \text{dom}(\mathsf{X}_{h'}).$$

No morphism in $\mathcal{C}_{\text{BAR}}^\partial$ may send a declared post-irreversible anchor class into the domain of a defined exact exchange operator unless the physical scope is first enlarged or an exchangeability preservation certificate is supplied inside the declared scope.

Proposition 5.3 ($\mathcal{C}_{\text{BAR}}^\partial$ is a category). *Partial baryogenesis reconstruction objects and domain-preserving partial reconstruction morphisms form a category.*

Proof. The identity morphism preserves E_{pre} , E_{post} , and $\text{dom}(\mathsf{X}_h)$. If Φ^∂ and Ψ^∂ are composable domain-preserving morphisms, their composition preserves each inclusion by transitivity. The underlying maps compose in $\mathcal{C}_{\text{BAR}}^{\text{pre}}$, and associativity is inherited from ordinary composition. Hence $\mathcal{C}_{\text{BAR}}^\partial$ is a category. $\square$

Remark 5.4 (Undefined exchange is not a third sign). Undefinedness of X_h on the post-irreversible anchor quotient is not a third matter type, not ignorance, and not a hidden anti-domain. It is non-membership in the admissible operator domain. The post-irreversible carrier still has material-reference content; what is absent is a lawful exact exchange operation on that content.

Theorem 5.5 (Exchange totalization obstruction). *No admissible same-scope extension totalizes X_h on the post-irreversible anchor quotient after genuine retrieval-irreversibility. Equivalently, there is no admissible same-scope operator*

$$\mathsf{X}_h^* : X_h \rightarrow X_h$$

that extends X_h and makes exact matter–antimatter exchange a globally defined late-time material-reference symmetry while preserving the same anchor quotient, role profiles, continuation relation, and retrieval status.

Proof. A totalized exchange operator X_h^* would make an undefined same-scope operation defined on the declared post-irreversible carrier. In AASC partial-domain closure, admissible extensions preserve the declared operator domains on the declared carrier: making an undefined same-scope operation defined is a repair or generator of admissibility, not an extension of the same envelope. If a certificate exists from the late scope's admitted data, then the operation was already in $E_{\text{post}}^{\text{cert}}$ and the history has not instantiated the relevant retrieval-irreversibility. If no certificate exists, totalization requires hidden global indexing, selector import, post-hoc reindexing, or scope change. Therefore X_h cannot be totalized on the declared post-irreversible anchor quotient. $\square$

Definition 5.6 (Internal and burden-restricted exchangeability obstructions). For a partial reconstruction object B^∂ , define the internal exchangeability obstruction

$$\mathcal{O}_{\text{ex}}(B^\partial)$$

as the obstruction, internal to $\mathbb{C}_{\text{BAR}}^\partial$, to the attempted totalization

$$X_h : E_{\text{pre}} \rightarrow X_h \quad \rightsquigarrow \quad X_h^* : X_h \rightarrow X_h.$$

Thus $\mathcal{O}_{\text{ex}}(B^\partial) = 0$ only if there exists an admissible same-scope totalization preserving the anchor quotient, role profiles, continuation relation, and retrieval status.

The *burden-restricted exchangeability obstruction*

$$\mathcal{O}_{\text{ex}}^{\text{bur}}(B^\partial)$$

is the corresponding obstruction to exact exchange-totalization on the burden-realizing post-irreversible anchor subquotient $\mathcal{Q}_S^{\text{bur}}(h)$. It is nonzero when exact exchangeability fails to totalize somewhere on the support that realizes the declared material-reference burden while preserving role profiles, continuation, retrieval status, and the same declared burden.

The *burden-wide exchange obstruction*

$$\mathcal{O}_{\text{ex}}^{\forall \text{bur}}(B^\partial)$$

holds when every member of every minimum burden-realizing subfamily $U \subseteq \mathcal{Q}_S^{\text{bur}}(h)$ lies outside the admissible domain of exact post-irreversible exchange transport preserving its role profile into $\mathcal{R}_{\text{bur}}$.

Remark 5.7 (Status of the obstruction objects). The symbols $\mathcal{O}_{\text{ex}}$, $\mathcal{O}_{\text{ex}}^{\text{bur}}$, and $\mathcal{O}_{\text{ex}}^{\forall \text{bur}}$ record three different statuses. The first records failure of full-carrier exchange totalization. The second records some exchange-totalization failure on the burden-realizing support. The third records burden-wide failure across every anchor class required by a minimum-cardinality realization of the declared burden. None is asserted to be a cohomological, K -theoretic, or standard cosmological invariant.

Theorem 5.8 (Global exchange-totalization obstruction). *For every stable coherent material-reference reconstruction object undergoing genuine retrieval-irreversibility and realizing persistent material reference,*

$$\mathcal{O}_{\text{ex}}(B^\partial) \neq 0.$$

Proof. If $\mathcal{O}_{\text{ex}}(B^\partial) = 0$, then an admissible same-scope totalization X_h^* exists on X_h . This contradicts the exchange totalization obstruction under genuine retrieval-irreversibility. Equivalently, the alleged

vanishing witness must either supply an exchangeability preservation certificate, import hidden data, repair the declared domain, introduce a selector, or change the physical scope. The first denies genuine retrieval-irreversibility; the remaining routes are inadmissible or larger-scope. $\square$

Proposition 5.9 (Burden-restricted obstruction criterion). *If exact exchange totalization fails somewhere on the realized burden support $\mathcal{Q}_S^{\text{bur}}(h)$ while preserving role profiles, continuation relation, retrieval status, and the same declared material-reference burden, then*

$$\mathcal{O}_{\text{ex}}^{\text{bur}}(B^\partial) \neq 0.$$

Proof. The burden-restricted obstruction is the restriction of exact exchange-totalization failure to $\mathcal{Q}_S^{\text{bur}}(h)$. Failure at any burden-realizing anchor class while preserving the protected burden data is precisely nonvanishing of $\mathcal{O}_{\text{ex}}^{\text{bur}}(B^\partial)$. This is a weak obstruction: it marks at least one burden-relevant failure and does not by itself supply the scope-indexed lower bound. $\square$

Proposition 5.10 (Burden-wide obstruction criterion). *If the declared burden is burden-wide genuinely retrieval-irreversible on $\mathcal{Q}_S^{\text{bur}}(h)$, then the burden-wide exchange obstruction $\mathcal{O}_{\text{ex}}^{\text{vbur}}(B^\partial)$ holds.*

Proof. Burden-wide genuine retrieval-irreversibility says that every anchor class in every minimum burden-realizing subfamily lacks the exchangeability preservation certificate required for exact post-irreversible transport of its role profile into $\mathcal{R}_{\text{bur}}$. That condition is exactly the burden-wide obstruction $\mathcal{O}_{\text{ex}}^{\text{vbur}}(B^\partial)$. $\square$

Lemma 5.11 (Burden-restricted obstruction-to-witness registration). *The nonvanishing obstruction $\mathcal{O}_{\text{ex}}^{\text{bur}}(B^\partial) \neq 0$ registers the witness predicate on the burden-realizing support $\mathcal{Q}_S^{\text{bur}}(h)$: a burden-realizing anchor class is witness-relevant exactly when its exchanged counterpart cannot be transported through the history while preserving the same downstream material-reference role profile. This registration does not by itself assert a scope-indexed lower bound. It gives witness nonemptiness when the burden support is nonempty; the lower bound requires the burden-wide obstruction $\mathcal{O}_{\text{ex}}^{\text{vbur}}(B^\partial)$.*

Proof. The burden-restricted obstruction states that exact exchange cannot be totalized at least somewhere on $\mathcal{Q}_S^{\text{bur}}(h)$ while preserving the anchor quotient, role profiles, continuation relation, retrieval status, and declared burden. The burden-realizing anchor classes at which the exchanged counterpart lacks such transportability are precisely the classes marked by the witness predicate. If the declared scope imposed no persistent material-reference burden, the support could be empty. When the support is nonempty, weak extraction supplies witness nonemptiness. The lower-bound extraction requires burden-wide failure over every minimum burden-realizing subfamily. $\square$

5.1 Witness-preserving audit category after obstruction

Definition 5.12 (Witness-preserving audit category). Let $\mathbf{C}_{\text{BAR}}^{\text{wit}}$ be the refinement of $\mathbf{C}_{\text{BAR}}^{\text{pre}}$ used after obstruction-to-witness extraction has registered the witness predicate and established nonempty burden-restricted witness support where $\mathcal{O}_{\text{ex}}^{\text{bur}}(B^\partial) \neq 0$. Objects are tuples

$$B^{\text{wit}} = (B^{\text{pre}}, \mathcal{W}_S(h), \text{arank}_S(h), \Delta_B^S(h)).$$

Morphisms in $\mathbf{C}_{\text{BAR}}^{\text{wit}}$ are pre-witness admissible morphisms that additionally preserve $\mathcal{W}_S(h)$, $\text{arank}_S(h)$, and the signed coordinate $\Delta_B^S(h)$ as audit data. The signed coordinate is not treated as

sign-fixed merely by entering $C_{\text{BAR}}^{\text{wit}}$; on SCMS scopes, sign-fixation is established only by the later sign-fixation theorem.

Definition 5.13 (Witness-preserving admissible reconstruction equivalence). Two witness-stage reconstruction objects $B^{\text{wit}}, B'^{\text{wit}} \in \text{Obj}(C_{\text{BAR}}^{\text{wit}})$ are *witness-admissibly equivalent*, written $B^{\text{wit}} \sim_{\text{adm}} B'^{\text{wit}}$, iff there exist admissible reconstruction morphisms in both directions preserving anchor quotients, role profiles, continuation, retrieval status, and persistent anchor-witness data. The uniqueness claims in this register concern the post-extraction audit quotient, not the pre-witness proof of obstruction or witness-predicate applicability.

Proposition 5.14 ($C_{\text{BAR}}^{\text{wit}}$ is a category). *Witness-stage baryogenesis reconstruction objects together with witness-preserving admissible reconstruction morphisms form a category.*

Proof. Identity maps preserve the pre-witness data and the audit witness data. Composition preserves the pre-witness data by the category property of $C_{\text{BAR}}^{\text{pre}}$, and it preserves witness families, anchor rank, and the signed coordinate by transitivity. Associativity is inherited from ordinary map composition. Hence $C_{\text{BAR}}^{\text{wit}}$ is a category. $\square$

Definition 5.15 (Persistent-carrier witness category and reconstruction functor). Let W_{BAR} be the category whose objects are persistent-carrier witness families

$$W_B = (\mathcal{Q}_S(h), \text{Role}_h^S, \text{Cont}_h, \mathcal{W}_S(h), \Delta_B^S(h), \text{Rec}_{\Xi}(h)),$$

where $\text{Rec}_{\Xi}(h)$ records whether the exchangeability preservation certificate exists. Morphisms in W_{BAR} are witness-preserving relabelings and chart changes preserving all displayed data. Define the reconstruction functor

$$F_{\text{BAR}} : C_{\text{BAR}}^{\text{wit}} \longrightarrow W_{\text{BAR}}$$

by sending B^{wit} to its persistent-carrier witness family and sending an admissible morphism to the induced map on the displayed witnesses.

Proposition 5.16 (Functoriality of persistent-carrier reconstruction). *The assignment $F_{\text{BAR}} : C_{\text{BAR}}^{\text{wit}} \rightarrow W_{\text{BAR}}$ is a functor. It sends identities to identities and composition of admissible reconstruction morphisms to composition of witness-preserving maps.*

Proof. The identity morphism on B^{wit} induces the identity map on W_B . If Φ and Ψ are composable admissible morphisms, then the induced witness map of $\Psi \circ \Phi$ is exactly the composition of the induced witness maps because the anchor quotient, role profile, continuation relation, retrieval status, and surplus witness are all preserved componentwise. Hence F_{BAR} is functorial. $\square$

Definition 5.17 (Universal witness-preserving persistent-carrier quotient). Let

$$\pi_{\text{BAR}} : C_{\text{BAR}}^{\text{wit}} \longrightarrow Q_{\text{BAR}}$$

be the quotient functor identifying exactly those witness-stage reconstruction objects and morphisms related by witness-admissible equivalence. The category Q_{BAR} is the universal admissible persistent-carrier quotient of the declared baryogenesis reconstruction scope at the witness-audit stage, after obstruction-to-witness extraction has registered the witness predicate and established nonempty burden-restricted support where the burden-restricted obstruction applies.

Proposition 5.18 (Categorical quotient universal property). *For any category $\mathcal{D}$ and any functor $G : \mathbb{C}_{\text{BAR}}^{\text{wit}} \rightarrow \mathcal{D}$ that is constant on witness-admissible equivalence classes of objects and morphisms, there exists a unique functor $\overline{G} : \mathbb{Q}_{\text{BAR}} \rightarrow \mathcal{D}$ such that*

$$G = \overline{G} \circ \pi_{\text{BAR}}.$$

Proof. Define $\overline{G}([B^{\text{wit}}]_{\text{adm}}) = G(B^{\text{wit}})$ and define $\overline{G}$ on morphism classes by applying G to any representative morphism. This is well-defined because G is constant on witness-admissible equivalence classes. The identity $G = \overline{G} \circ \pi_{\text{BAR}}$ follows by construction. Uniqueness follows because π_{BAR} is surjective onto quotient object and morphism classes. $\square$

Corollary 5.19 (Equivalence-class uniqueness at the witness-audit stage). *Within $\mathbb{C}_{\text{BAR}}^{\text{wit}}$, after obstruction-to-witness extraction has registered the witness predicate and established the relevant witness data for the audit stage, every witness-stage reconstruction preserving the same anchor quotient, role profile, retrieval status, continuation relation, and witness family factors through the same object of $\mathbb{Q}_{\text{BAR}}$. Full sign-fixed uniqueness is stated only after the SCMS sign-fixation theorem.*

Proof. By the quotient universal property, every admissible same-scope witness-stage reconstruction factors through the same quotient object determined by the anchor quotient, role profile, retrieval status, continuation relation, and registered persistent anchor-witness data. A second object preserving these data is the same quotient object. If it does not preserve them, it is not a same-scope witness-stage reconstruction. Sign-side uniqueness requires SCMS connectivity and is established by the sign-fixation result. $\square$

Theorem 5.20 (Conservative reconstruction collapse for hidden exchange sectors). *Let B^* be an admissibility-preserving extension of a baryogenesis reconstruction object that adds a hidden exchange-sector symbol or historical index while preserving all declared anchor quotients, material-reference roles, continuation relations, and surplus witnesses. Then B^* is definitional/bookkeeping-equivalent to the prior reconstruction on the declared scope. If it is not bookkeeping-equivalent, it changes witnesses, introduces a selector, repairs an undefined exchange assignment, imports a carrier, or changes scope.*

Proof. Conservativity on the declared scope means that every prior standing-bearing witness is unchanged. A new symbol that changes no prior witness is skin unless it is explicitly definable from the declared anchor quotient and continuation structure. If the symbol is not definable from the prior witnesses but is treated as standing-bearing, its authority must come from a hidden selector, repaired exchange totalization, carrier laundering, or scope transport. Those are inadmissible or larger-domain moves. Hence a conservative hidden exchange sector collapses to bookkeeping. $\square$

Definition 5.21 (Baryogenesis reconstruction obstruction classes). Every same-scope attempt to escape persistent anchor-witness reconstruction has an image in the following obstruction family. The sign-fixed version is the SCMS specialization established in Section 11:

Class	Reconstruction escape route
B1	Selector insertion: a hidden global index, canonical pairing, typicality measure, preferred representative, or sign selector chooses an exchange continuation not fixed by the anchor quotient.
B2	Reindexing repair/generator: an undefined post-irreversible exchange operation is made defined after the fact, or a lost exchange certificate is generated from an ineligible late history.
B3	Carrier laundering: a derived quantity, symmetry slogan, entropy story, mechanism label, probability, or explanatory virtue is treated as primitive authorization for exact exchange transport.
B4	Scope transport: detector/record structure, interaction channels, domain decomposition, cosmological boundary conditions, or material-reference burden are changed while the result is presented as the same reconstruction problem.
B5	Witness erasure: the persistent material-reference burden is weakened until no anchor witness remains, while the resulting history is still advertised as a same-scope persistent-material history.
B6	Bookkeeping duplication: a hidden exchange label preserves all witnesses and continuation data, adding no standing-bearing distinction.

Theorem 5.22 (Reconstruction obstruction classification). *Every same-scope attempt to avoid persistent anchor-witness reconstruction while preserving the declared persistent material-reference burden instantiates at least one of the obstruction classes B1–B6. Classes B1–B5 are inadmissible or larger-scope; class B6 is inert bookkeeping. Therefore no further same-scope reconstruction route remains.*

Proof. An escape attempt must acquire force by selecting an exchange continuation, repairing lost exchange definedness, authorizing a hidden carrier, changing scope, erasing the witness burden, or duplicating prior witnesses without effect. These are exactly B1–B6. B1 violates no-selector discipline. B2 violates partial-domain closure and irreversibility. B3 imports a carrier not fixed by the declared anchor quotient. B4 changes the theorem object. B5 exits the material-reference target. B6 changes no standing-bearing content and is skin. Since anchor/tensor/skin role decomposition leaves no fourth role and fixed-domain exhaustion leaves no fifth governance-bearing locus, no further same-scope route exists. Claims about later redescription or feedback amplification are handled separately as surplus-class redescription routes rather than exchangeability-preservation routes. $\square$

6 Boundary Regime, Dependency Order, and Fixed Admissible Surplus Class

The preceding sections define the anchor quotient, the pre-witness category, the exchange-totalization obstruction, and the witness-audit register. The boundary object can now be stated without formal prematurity: it is the dependency-minimal licensing status in which those reconstruction data are jointly available.

Definition 6.1 (Reconstruction-status dependency preorder). Let $\text{RecStat}_{\mathcal{S}}(h)$ be the class of licensed reconstruction packages available for the fixed material-reference problem $(\mathcal{S}, h)$. For $X, Y \in \text{RecStat}_{\mathcal{S}}(h)$, write

$$X \preceq_{\text{dep}} Y$$

when every standing-bearing datum licensed in X is required for the licensed use of Y , while Y may contain additional downstream mechanism, witness, parameter, or presentation data. The relation $\preceq_{\text{dep}}$ is a dependency preorder: it is reflexive, transitive, and quotientable by mutual dependency equivalence.

Lemma 6.2 (Dependency-order quotient). *Mutual dependency,*

$$X \equiv_{\text{dep}} Y \iff X \preceq_{\text{dep}} Y \text{ and } Y \preceq_{\text{dep}} X,$$

identifies reconstruction packages differing only by dependency-equivalent presentation. Therefore minimality claims are made on $\text{RecStat}_{\mathcal{S}}(h)/\equiv_{\text{dep}}$, not on arbitrary syntax.

Proof. Reflexivity and transitivity make $\preceq_{\text{dep}}$ a preorder. Mutual dependency identifies packages that license the same standing-bearing data for the same downstream uses, even if their syntax, names, or presentation differ. Quotienting by this equivalence gives the comparison carrier on which dependency-minimality is exact. $\square$

Definition 6.3 (Persistent-material boundary regime). The *persistent-material boundary regime* $\mathcal{B}_{\mathcal{S}}(h)$ is a $\preceq_{\text{dep}}$ -minimal licensed reconstruction package, up to $\equiv_{\text{dep}}$, containing the anchor quotient, burden-realizing support, role profile, continuation relation, retrieval status, persistent witness family, and anchor-rank coordinate:

$$\mathcal{Q}_{\mathcal{S}}(h), \quad \mathcal{Q}_{\mathcal{S}}^{\text{bur}}(h), \quad \text{Role}_h^{\mathcal{S}}, \quad \text{Cont}_h, \quad \text{Rec}_{\Xi}(h), \quad \mathcal{W}_{\mathcal{S}}(h), \quad \text{arank}_{\mathcal{S}}(h).$$

The boundary regime is a dependency-minimal licensing status, not a first temporal instant, causal event, thermodynamic epoch, phase transition, or baryogenesis mechanism.

Definition 6.4 (Boundary-fixed admissible surplus reconstruction classes). The unsigned admissible surplus reconstruction class of h on $\mathcal{S}$ is

$$\mathfrak{S}_{\mathcal{S}}^{\text{uns}}(h) := (\mathcal{Q}_{\mathcal{S}}(h), \mathcal{Q}_{\mathcal{S}}^{\text{bur}}(h), \text{Role}_h^{\mathcal{S}}, \text{Cont}_h, \text{Rec}_{\Xi}(h), \mathcal{W}_{\mathcal{S}}(h), \text{arank}_{\mathcal{S}}(h)) / \sim_{\text{adm}}.$$

On an SCMS scope, the signed specialization is

$$\mathfrak{S}_{\mathcal{S}}^{\text{SCMS}}(h) := (\mathfrak{S}_{\mathcal{S}}^{\text{uns}}(h), \Delta_B^{\mathcal{S}}(h)) / \sim_{\text{adm}}.$$

The signed coordinate is added only at the SCMS specialization stage; Section 11 proves that this sign is fixed. Neither class encodes a numerical particle abundance, a baryon-to-photon ratio, a thermal rate, a CP phase value, or a selected physical mechanism.

Proposition 6.5 (Conditional boundary fixation). *Conditional on the exchange-totalization obstruction, weak obstruction-to-witness extraction, burden-wide lower-bound extraction where the lower-bound hypothesis is assumed, symmetric-continuation exhaustion, and surplus-class amplification exhaustion proved in the theorem package, the unsigned reconstruction class $\mathfrak{S}_{\mathcal{S}}^{\text{uns}}(h)$ is fixed at $\mathcal{B}_{\mathcal{S}}(h)$. On SCMS scopes the signed specialization $\mathfrak{S}_{\mathcal{S}}^{\text{SCMS}}(h)$ is fixed after sign fixation.*

Proof. If the relevant surplus reconstruction class were not fixed at $\mathcal{B}_{\mathcal{S}}(h)$, the later material-reference burden would have to be supplied by hidden selector import, post-hoc repair, carrier laundering, repaired exchange totalization, witness erasure, scope transport, or redescription amplification.

The named obstruction and exhaustion theorems eliminate those routes as same-scope admissible reconstructions. The final boundary-fixation theorem combines these route closures after the obstruction, extraction, and exhaustion results have been established. $\square$

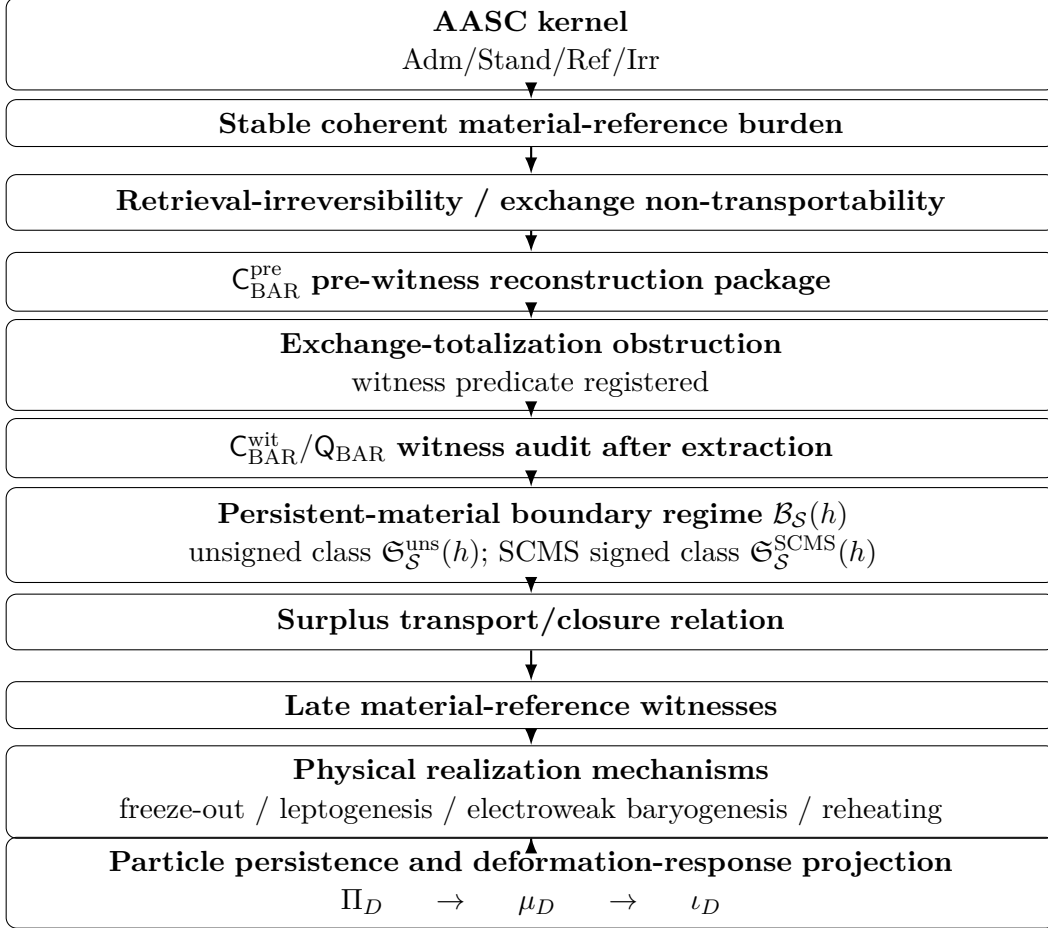


Figure 2: Dependency structure of the matter–antimatter reconstruction theorem. The arrows record reconstruction dependency, not temporal sequence. Physical baryogenesis mechanisms may realize projected particle-number asymmetry inside the boundary-fixed class; they do not generate the admissibility of that class by later redescription.

7 Surplus-Class Transport and Late Witness Closure

The boundary-fixed surplus reconstruction class is not merely declared and then abandoned. It must remain available for later material-reference use. A later baryonic, chemical, observational, or record-bearing witness is admissible only if its material-reference standing descends through lawful redescription from the boundary-fixed admissible surplus reconstruction class.

Definition 7.1 (Surplus transport/closure relation). A *surplus transport/closure relation* for $\mathcal{S}, h$ is a standing-preserving relation

$$\text{TC}_S^\Delta(\mathcal{B}_S(h), W_{\text{late}}^{\text{mat}})$$

from the boundary-fixed unsigned class $\mathfrak{S}_S^{\text{uns}}(h)$, and from $\mathfrak{S}_S^{\text{SCMS}}(h)$ on SCMS scopes, to the later material-reference witness family $W_{\text{late}}^{\text{mat}}$, preserving anchor quotient identity, burden-realizing support, downstream role profiles, retrieval-irreversibility status, admissible continuation, and no-selector/no-repair discipline; on SCMS scopes it also preserves sign-side fixation.

Proposition 7.2 (Late witness closure requirement). *If a later material-reference witness family is used as evidence of persistent cosmological identity, its standing-bearing coherence must be transported or closed from $\mathfrak{S}_S^{\text{uns}}(h)$. Independent local trajectory continuation is insufficient whenever it cannot preserve anchor quotient, role profile, retrieval status, and sign-side burden without hidden selector import, repair, or scope change.*

Proof. A late material-reference witness is not merely a pattern display. It is used as a witness of continuing material identity. Such use requires lawful lineage from the boundary-fixed reconstruction package to the late witness family. If local trajectory continuation preserves the anchor quotient, role profile, retrieval status, and sign-side burden, it supplies that lineage. If it does not, then an additional transport/closure relation must be declared, or the explanation collapses into hidden selector import, repair, or scope change. $\square$

8 Toy Histories, Freeze-Out Witness, and Transport Failure

Definition 8.1 (Pre-irreversible exchangeability). A history prefix $h|_{<t_*}$ is *pre-irreversibly exchangeable* if there exists an involution on its matter and antimatter carrier classes that swaps them while preserving every standing-bearing feature admitted by the prefix scope.

Definition 8.2 (Post-irreversible admissible transportability of exchange symmetry). A pre-irreversible exchange symmetry is *post-irreversibly admissibly transportable* if the same symmetry can be lawfully transported through the irreversible stage into the late-time anchor quotient while preserving sign type and downstream material-reference role profiles.

Definition 8.3 (Toy symmetric annihilation history). A toy symmetric annihilation history consists of:

1. an early stage with paired matter and antimatter carriers,
2. an annihilation stage in which paired carriers are dynamically removed, and
3. a dispersal stage in which the surviving radiation and residual structures no longer carry the information required for lawful global relabeling of the original pairs.

It is *toy-symmetric* if the early history is pre-irreversibly exchangeable and that symmetry is demanded to remain post-irreversibly admissibly transportable.

Definition 8.4 (Toy freeze-out witness). A *toy freeze-out witness* is a history fragment with four stages:

1. an early exchangeable plasma stage t_0 in which matter and antimatter carriers occur in paired exchange classes;
2. an interaction stage t_1 in which annihilation, scattering, and conversion processes erase pairwise carrier correlations;

3. a freeze-out/dispersal stage t_2 in which the correlation data required for a global exchange map is no longer retrievable on $\mathcal{S}$; and
4. a late persistence stage t_3 in which some surviving anchor class supports chemistry-bearing or observer-bearing material reference.

The witness is not a numerical cosmology. It is a minimal structural model of the transport problem.

Proposition 8.5 (Toy freeze-out failure certificate). *In a toy freeze-out witness, exact exchangeability can be harmless at t_0 but cannot be transported as exact late-time material-reference symmetry through t_2 into t_3 unless the late scope contains a lawful recovery map for the lost global indexing.*

Proof. At t_0 , the exchange map acts on available paired carrier data and preserves the early prefix. By t_2 , the data needed to identify all late carrier structures as images of that global exchange map has been lost on the fixed scope. At t_3 , material reference is carried by the late anchor quotient. Transporting the early exchange map to t_3 therefore requires a recovery map through data absent from $\mathcal{S}$. That is hidden load unless the map is explicitly present as a lawful same-scope witness. If it is not present, exact exchangeability fails to survive as late-time material-reference symmetry. $\square$

Lemma 8.6 (No-global-reindexing lemma). *Let $h \in \mathcal{H}$ exhibit irreversible loss of retrievable correlation on scope $\mathcal{S}$. Then no later same-history relabeling of matter and antimatter classes can restore a persistent material-reference structure that was not already fixed before the loss.*

Proof. Suppose such a relabeling were possible. Then after irreversible loss, one could still define a lawful map sending the late-time carrier structure back to a globally coherent earlier matter–antimatter indexing. But this is exactly what the irreversibility condition excludes: once the information needed for that map has been irretrievably lost on the fixed scope, any such reconstruction depends on extra information not carried by the history. It is therefore hidden load or scope change, not admissible same-history relabeling. $\square$

Proposition 8.7 (Pre-irreversible exchangeability is not itself the failure). *Pre-irreversible exchangeability is not, by itself, inadmissible. The contradiction appears only when the same symmetry is demanded to remain post-irreversibly admissibly transportable through a stage of irreversible loss.*

Proof. Before the irreversible stage, the exchange involution acts on a history prefix whose matter–antimatter correlations remain retrievable. On that prefix the symmetry is a lawful bookkeeping symmetry. The obstruction appears only when the same involution must be transported into the late anchor quotient after the requisite correlations have been lost. By the no-global-reindexing lemma, that transport is not available. $\square$

9 Unified Exchangeability Collapse and No Exact Symmetric Persistent Continuation

The previous sections separate three notions that are often conflated: label exchange, carrier-role exchange, and lawful transport of an exchange symmetry through irreversible history. This section states the collapse theorem in a single place.

Definition 9.1 (Exchangeability preservation certificate). An *exchangeability preservation certificate* for h on $\mathcal{S}$ is a same-scope admissible recoverability certificate that carries a pre-irreversible matter–antimatter exchange involution into the post-irreversible anchor quotient while preserving:

1. the carrier-role profile of every standing-bearing late anchor class,
2. the downstream material-reference role map,
3. the admissible continuation relation, and
4. the declared material-reference burden of the scope.

Lemma 9.2 (Genuine retrieval-irreversibility / certificate equivalence). *On a fixed material-reference scope $\mathcal{S}$, a history h exhibits genuine retrieval-irreversibility for the matter–antimatter exchange problem iff no exchangeability preservation certificate is constructible from the late admitted data and lawful redescription rules of $\mathcal{S}$.*

Proof. If such a certificate exists, the pre-irreversible exchange involution can be carried into the post-irreversible anchor quotient while preserving role profiles, continuation, retrieval status, and material-reference burden. The relevant exchange data are therefore still admissibly recoverable on $\mathcal{S}$, so genuine retrieval-irreversibility has not occurred. Conversely, if genuine retrieval-irreversibility fails for the exchange problem, then the late admitted data and redescription rules contain a lawful recovery route for the exchange involution; that route is exactly an exchangeability preservation certificate. Hence genuine retrieval-irreversibility and certificate absence coincide for the fixed exchange problem. $\square$

This equivalence is scope-relative. It does not deny that a larger theoretical description may retain formal microstate information; it says that such information is not a same-scope exchangeability certificate unless it is admissibly available in $\mathcal{S}$.

Lemma 9.3 (Exchange-relevant retrieval-irreversibility lemma). *Let h undergo irreversible loss of retrievable correlation on $\mathcal{S}$. If the late scope does not contain the data and lawful redescription rules required to construct an exchangeability preservation certificate, then no exact matter–antimatter exchange map is recoverable as a same-scope identity symmetry. In particular, global microstate retention is irrelevant to the theorem unless it induces such a certificate from the data admitted by $\mathcal{S}$.*

Proof. Recovering exact exchangeability as a same-scope identity symmetry requires more than an external statement that some total description contains correlations. It requires a lawful map, available from the late scope’s admitted data and redescription rules, that carries the pre-irreversible exchange involution into the post-irreversible anchor quotient while preserving carrier-role profiles, downstream material-reference roles, admissible continuation, and the declared material-reference burden of the scope. That map is exactly an exchangeability preservation certificate. If the late scope lacks the data or rules needed to construct it, then any alleged recovery uses hidden external indexing, a selector, or a changed theorem object rather than same-scope admissible recovery. Hence exact exchangeability is not recoverable as a same-scope identity symmetry. $\square$

Theorem 9.4 (Exchangeability transport failure theorem). *Let h undergo irreversible loss of admissibly recoverable correlation on $\mathcal{S}$. If no exchangeability preservation certificate exists after $t_*(h)$, then exact matter–antimatter exchangeability is not post-irreversibly transportable as a same-history identity symmetry.*

Proof. Post-irreversible transportability of exact exchangeability requires a lawful map carrying the early exchange involution into the late anchor quotient without changing carrier-role profiles or admissible continuation. That is precisely an exchangeability preservation certificate. If no such certificate exists, then any attempted exchange either changes carrier-role content, imports a selector, or performs post-hoc reindexing. None is a same-history identity symmetry on the fixed scope. $\square$

Theorem 9.5 (Unified exchangeability-collapse theorem). *On a fixed cosmological scope instantiating stable coherent material reference, every attempted post-irreversible preservation of exact matter–antimatter exchangeability falls into exactly one of three classes:*

1. *skin exchange: it preserves carrier-role profiles, the anchor quotient, and admissible continuation, and therefore carries no standing-bearing asymmetry content;*
2. *certified transport: it supplies an exchangeability preservation certificate and so has not undergone the relevant retrieval-irreversibility on that scope;*
3. *carrier-role motion: it changes carrier-role content, witness structure, sign side, or continuation without a certificate, and therefore collapses into hidden selector import, post-hoc reindexing, standing failure, or scope change.*

Consequently, after genuine retrieval-irreversibility, no standing-bearing exact exchangeability remains on the same scope.

Proof. An attempted exchange either changes the carrier-role profile or it does not. If it does not, then by carrier-role non-equivalence it is skin and belongs to class (1). If it does change carrier-role content, then it must be supported by a lawful transport map from the pre-irreversible exchange symmetry into the late anchor quotient. If such a map exists, it is an exchangeability preservation certificate and class (2) holds; in that case the relevant exchange information remains admissibly recoverable on the scope. If no such map exists, the carrier-role change is supplied by a selector, post-hoc relabeling, hidden historical indexing, or a changed theorem object. That is class (3). These three cases are exhaustive by bivalence of the standing-bearing distinction: the exchange either changes standing-bearing carrier-role content or it does not; if it does, the required transport certificate either exists or does not. $\square$

Theorem 9.6 (No exact symmetric persistent continuation theorem). *Let $\mathcal{S}$ be a fixed material-reference scope and let $h \in \mathcal{H}$ undergo irreversible loss of admissibly recoverable correlation. If h remains admissible on $\mathcal{S}$ and realizes persistent material reference after $t_*(h)$, then h cannot also realize exact matter–antimatter symmetric persistent continuation on the same scope.*

Proof. Persistent material reference after $t_*(h)$ requires late carrier-role profiles to be fixed in the anchor quotient. Exact symmetric persistent continuation would require post-irreversible transport of the matter–antimatter exchange symmetry while preserving that same quotient and continuation relation. By the exchangeability transport failure theorem, this requires an exchangeability preservation certificate. But h has undergone irreversible loss of admissibly recoverable correlation, so no such same-scope certificate exists. The unified exchangeability-collapse theorem then leaves only skin exchange or carrier-role motion. Skin exchange is not standing-bearing exact symmetric continuation; carrier-role motion is hidden selector import, post-hoc reindexing, standing failure, or scope change. Hence exact symmetric persistent continuation is impossible on the fixed scope. $\square$

Corollary 9.7 (Main non-survival form). *For the target class of histories considered in this paper, exact global matter–antimatter exchangeability is not a late-time admissible symmetry of persistent material-reference structure. It is either pre-irreversible bookkeeping, non-standing skin, or a failed transport claim.*

10 Anchor Realization, Symmetric Identity Failure, and Lower Bounds

Lemma 10.1 (Anchor realization). *Every standing-bearing material reference class in scope $\mathcal{S}$ requires realization by persistent anchor content in the post-irreversible witness structure. In particular, the number of distinct persistent anchor classes realizing the declared material-reference burden is at least $\rho_{\mathcal{S}}$.*

Proof. Let $r \in \mathcal{R}$ be a standing-bearing material reference class after irreversibility. By definition of persistent material reference, r must remain attached to some surviving late-time material structure under lawful continuation. Hence some persistent anchor class must realize it. The number $\rho_{\mathcal{S}}$ is defined as the minimum number of distinct persistent anchor classes required to realize all such reference classes on the fixed scope. Therefore any admissible realization of the declared burden contains at least $\rho_{\mathcal{S}}$ distinct persistent anchor classes. $\square$

Lemma 10.2 (Weak obstruction-to-witness extraction). *If $\mathcal{O}_{\text{ex}}^{\text{bur}}(B^\partial) \neq 0$ and $\mathcal{Q}_{\mathcal{S}}^{\text{bur}}(h) \neq \emptyset$, then at least one burden-realizing anchor class is witness-relevant. In particular,*

$$\mathcal{W}_{\mathcal{S}}^{\text{bur}}(h) \neq \emptyset, \quad \text{arank}_{\mathcal{S}}^{\text{bur}}(h) \geq 1.$$

Proof. The obstruction $\mathcal{O}_{\text{ex}}^{\text{bur}}(B^\partial) \neq 0$ states that exact exchange totalization fails at some point of the burden-realizing support while preserving role profiles, continuation relation, retrieval status, and the declared burden. A burden-realizing anchor class at which the exchanged counterpart is not admissibly transportable is exactly witness-relevant by the candidate anchor-witness definition. Since $\mathcal{Q}_{\mathcal{S}}^{\text{bur}}(h)$ is nonempty, at least one such class exists in the burden-restricted witness family. This proves nonemptiness and the weak rank bound $\text{arank}_{\mathcal{S}}^{\text{bur}}(h) \geq 1$. $\square$

The lower-bound theorem is not a consequence of weak burden-restricted obstruction alone; it uses the burden-wide hypothesis that every member of every minimum burden-realizing subfamily lacks admissible exact post-irreversible exchange transport.

Lemma 10.3 (Burden-wide lower-bound extraction). *If $\mathcal{O}_{\text{ex}}^{\text{vbur}}(B^\partial)$ holds, $\rho_{\mathcal{S}} \geq 1$, and anchor realization holds for the declared material-reference burden $\mathcal{R}_{\text{bur}}$, then*

$$\text{arank}_{\mathcal{S}}^{\text{bur}}(h) \geq \rho_{\mathcal{S}}, \quad \text{arank}_{\mathcal{S}}(h) \geq \text{arank}_{\mathcal{S}}^{\text{bur}}(h) \geq \rho_{\mathcal{S}}.$$

Proof. By anchor realization, there is a burden-realizing subfamily $U \subseteq \mathcal{Q}_{\mathcal{S}}^{\text{bur}}(h)$ for $\mathcal{R}_{\text{bur}}$. By the definition of $\rho_{\mathcal{S}}$, every burden-realizing subfamily has cardinality at least $\rho_{\mathcal{S}}$, and some minimum burden-realizing family has cardinality $\rho_{\mathcal{S}}$ when the declared burden is realized. The burden-wide obstruction $\mathcal{O}_{\text{ex}}^{\text{vbur}}(B^\partial)$ says that every member of every minimum burden-realizing subfamily lacks admissible exact post-irreversible exchange transport preserving its role profile into $\mathcal{R}_{\text{bur}}$. Hence every member of any minimum burden-realizing family of cardinality $\rho_{\mathcal{S}}$ is witness-relevant.

Therefore the burden-restricted witness family has at least ρ_S members. The unrestricted rank includes the burden-restricted witness family, so $\text{arank}_S(h) \geq \text{arank}_S^{\text{bur}}(h) \geq \rho_S$. $\square$

Theorem 10.4 (Symmetric identity failure theorem). *Let $h \in \mathcal{H}$ be a history on scope S . If:*

1. *S requires persistent material reference,*
2. *h undergoes irreversible loss of retrievable correlation, and*
3. *a pre-irreversible matter–antimatter exchange symmetry is demanded to remain post-irreversibly admissibly transportable,*

then h is inadmissible on the fixed scope.

Proof. Because the scope requires persistent material reference, the anchor realization lemma implies that the late history must contain persistent anchor classes realizing the standing-bearing material references of the scope. Because anchor equivalence is canonical up to unique quotient isomorphism, these classes are fixed by sign type and downstream role profile.

If the pre-irreversible exchange symmetry remained post-irreversibly admissibly transportable, then the late history would admit a lawful involution identifying realized anchor classes with exchanged partners while preserving the same downstream role profile. But that involution must either:

1. identify late anchor classes without preserving the sign-sensitive anchor quotient, collapsing the very anchor content required by realization; or
2. change sign type only by a global relabeling of matter and antimatter after irreversibility.

The second route is excluded by the no-global-reindexing lemma. The first collapses the anchor quotient needed to realize the scope’s persistent material-reference burden. Therefore the history is inadmissible on the fixed scope. $\square$

Corollary 10.5 (Nonzero anchor-rank necessity). *If a history h is admissible on scope S , supports persistent material reference through irreversible evolution, and satisfies $\mathcal{O}_{\text{ex}}^{\text{bur}}(B^\partial) \neq 0$ on a nonempty burden-realizing support, then*

$$\text{arank}_S(h) \geq 1.$$

Exact symmetric persistence is therefore excluded on any chemistry-bearing or observer-bearing scope whose declared burden support is exchange-obstructed.

Proof. The weak obstruction-to-witness extraction lemma gives $\mathcal{W}_S^{\text{bur}}(h) \neq \emptyset$ and $\text{arank}_S^{\text{bur}}(h) \geq 1$. Since burden-restricted witnesses are included in the total witness family, $\text{arank}_S(h) \geq 1$. $\square$

The following theorem uses the burden-wide hypothesis. It is not a consequence of weak burden-restricted obstruction alone.

Theorem 10.6 (Scope-indexed lower-bound theorem). *Let $h \in \mathcal{H}$ be admissible on scope S , realize the declared burden $\mathcal{R}_{\text{bur}}$, and satisfy burden-wide genuine retrieval-irreversibility on $\mathcal{Q}_S^{\text{bur}}(h)$, equivalently $\mathcal{O}_{\text{ex}}^{\text{vbur}}(B^\partial)$. Then*

$$\text{arank}_S(h) \geq \rho_S.$$

Equivalently, under burden-wide exchange non-transportability, the realized late-time anchor quotient contains at least as many persistent witness classes as the declared burden requires.

Proof. By definition, ρ_S is the minimum number of distinct persistent anchor classes in $\mathcal{Q}_S^{\text{bur}}(h)$ required to realize the declared burden $\mathcal{R}_{\text{bur}}$ after irreversible evolution. Burden-wide genuine retrieval-irreversibility gives $\mathcal{O}_{\text{ex}}^{\text{bur}}(B^\partial)$. The burden-wide lower-bound extraction lemma gives $\text{arank}_S^{\text{bur}}(h) \geq \rho_S$, and every burden-restricted witness is part of the unrestricted witness family. Hence $\text{arank}_S(h) \geq \text{arank}_S^{\text{bur}}(h) \geq \rho_S$. $\square$

This is a structural admissibility lower bound on persistent carrier-role witnesses, not a numerical baryon-asymmetry prediction and not a phenomenological abundance calculation.

Theorem 10.7 (Structural non-derivability of the exact asymmetry value). *Within the present reconstruction load, no exact numerical value of the observational asymmetry parameter is internally derivable. Any such value claim requires additional dynamical witnesses, a privileged selector or measure, or a scope change.*

Proof. The reconstruction theorem determines only burden-support witness existence, SCMS sign fixation, and the minimum anchor burden of the admissible surplus reconstruction class, represented by a persistent anchor witness. It does not introduce Boltzmann equations, nonequilibrium transport coefficients, CP -violating parameters, sphaleron rates, reheating data, or a measure selecting one numerical realization from another. Any exact numerical value for the observational asymmetry therefore requires additional dynamical witness structure, a privileged choice principle not fixed by the present scope, or a stronger scope in which such physical data are admitted. None belong to this reconstruction load. $\square$

11 Persistent-Class Conservation, SCMS Connectivity, and No Late Re-symmetrization

Remark 11.1 (SCMS connectivity). SCMS scope was defined in Definition 2.12. This section uses that connected sign-side burden to prove sign fixation. SCMS connectivity is not an extra baryogenesis mechanism: it records the ordinary burden imposed by a connected chemistry-bearing or observer-bearing material organization. A cosmology with disconnected matter and antimatter domains can be analyzed by indexed scopes; it is not a same-scope counterexample to a SCMS theorem.

Lemma 11.2 (Persistent-class conservation). *Let $h \in \mathcal{H}$ be admissible on scope $\mathcal{S}$. A persistent anchor witness set cannot reverse sign on the same admissible lineage without either illicit global reindexing or scope change.*

Proof. Assume a history segment in which the witness set is nonzero and matter-side, but later along the same admissible lineage the witness set is antimatter-side. Then either the earlier matter-side anchor classes are identified with later antimatter-side classes by a global relabeling of matter and antimatter, or the later stage changes which sign class carries the material-reference burden. The first route is excluded by the no-global-reindexing lemma after irreversibility. The second changes the fixed scope by altering the standing-bearing burden. Hence sign reversal is unavailable on the same admissible lineage. $\square$

Lemma 11.3 (Matter-side role-fixation theorem). *Let $h \in \mathcal{H}$ be admissible on scope $\mathcal{S}$, and suppose that at some stage $t_0 \geq t_*(h)$ the witness set is nonzero. Then the sign side carrying the downstream*

material-reference role is fixed up to lawful redescription for all later stages of the same admissible lineage.

Proof. At stage t_0 , the nonzero witness set realizes the standing-bearing material reference burden through anchor classes on one sign side. Because anchor equivalence is fixed by sign type and full downstream role profile, lawful redescription may reorganize representatives within that sign side but may not migrate the realized role to the opposite sign side without either:

1. identifying opposite-sign anchor classes as the same anchor class, contrary to the sign-sensitive anchor quotient, or
2. performing a global reindexing of matter and antimatter labels after irreversibility.

The second route is excluded by the no-global-reindexing lemma. The first would collapse anchor identity itself. Therefore the matter-bearing role is fixed up to lawful redescription once the nonzero witness set is realized. $\square$

Theorem 11.4 (Sign-fixation theorem). *Let $h \in \mathcal{H}$ be admissible on a sign-coherent material-standing scope $\mathcal{S}$, and suppose there exists a first stage at which the anchor witness set is nonzero. Then the sign of $\Delta_B^{\mathcal{S}}(h)$ is fixed along the same admissible lineage.*

Proof. By matter-side role fixation, the sign side carrying the material-reference role is fixed once the nonzero witness set is realized. By SCMS connectivity, the standing-bearing material reference burden cannot be split into mutually annihilative opposite-sign anchor lineages while remaining one connected material organization on the same scope. By persistent-class conservation, the sign cannot reverse later without illicit reindexing or scope change. Therefore the sign of $\Delta_B^{\mathcal{S}}(h)$ is fixed along the admissible lineage. $\square$

Corollary 11.5 (Signed lower bound after sign fixation). *If $h \in \mathcal{H}$ is admissible on a sign-coherent material-standing scope $\mathcal{S}$, its nonzero witness set has been realized, and burden-wide exchange non-transportability holds on $\mathcal{Q}_S^{\text{bur}}(h)$, then*

$$|\Delta_B^{\mathcal{S}}(h)| = \text{arank}_{\mathcal{S}}(h) \geq \rho_{\mathcal{S}}.$$

Proof. After sign fixation on a sign-coherent material-standing scope, exactly one of $\mathcal{W}_S^{\text{M}}(h)$ or $\mathcal{W}_S^{\bar{\text{M}}}(h)$ carries the connected downstream material-reference role. Therefore

$$|\Delta_B^{\mathcal{S}}(h)| = ||\mathcal{W}_S^{\text{M}}(h)| - |\mathcal{W}_S^{\bar{\text{M}}}(h)|| = |\mathcal{W}_S^{\text{M}}(h)| + |\mathcal{W}_S^{\bar{\text{M}}}(h)| = \text{arank}_{\mathcal{S}}(h).$$

The lower bound then follows from the scope-indexed lower-bound theorem. $\square$

Theorem 11.6 (No late re-symmetrization theorem). *Let $h \in \mathcal{H}$ be admissible on a sign-coherent material-standing scope $\mathcal{S}$. Once the witness sign has been fixed and $\Delta_B^{\mathcal{S}}(h) \neq 0$, no later same-history stage may satisfy post-irreversible admissible transportability of exact matter–antimatter exchangeability without standing failure or scope change.*

Proof. A later exactly exchangeable stage would have to erase the already fixed sign side of the witness quotient by reintroducing a lawful late-time transport of matter–antimatter symmetry through the irreversible stage. But by the sign-fixation theorem and matter-side role fixation, that

sign side is fixed on the same admissible lineage. Thus exact late re-symmetrization either collapses standing-bearing anchor identity, imports a global reindexing excluded by the no-global-reindexing lemma, or changes the scope. It is not admissible on the same scope. $\square$

Corollary 11.7 (Universal quotient form after sign fixation). *Within the witness-preserving quotient on an SCMS scope, after obstruction-to-witness extraction registers the witness predicate, weak extraction establishes nonempty burden-restricted support where applicable, and the sign-fixation theorem applies, the sign-fixed persistent anchor-witness reconstruction is unique up to admissible bookkeeping equivalence. An alleged second same-scope object either maps to the same object of $\mathbf{Q}_{\text{BAR}}$, violates domain preservation, instantiates a reconstruction obstruction route, or changes physical scope.*

Proof. Witness-stage uniqueness follows from the quotient universal property in $\mathbf{C}_{\text{BAR}}^{\text{wit}}$. The sign-fixation theorem identifies the sign side carrying the connected material-reference burden on an SCMS scope. Therefore two witness-stage reconstructions preserving anchor quotient, role profiles, continuation, retrieval status, and the sign-fixed witness coordinate are the same admissible quotient object. A rival that changes one of these data is not a same-scope witness-stage reconstruction. $\square$

12 Symmetric-Continuation Route Exhaustion

The previous sections prove that exact exchangeability cannot be transported through irreversible history once persistent material reference is required. This section closes the exact-exchange escape space. The route labels are generated by the exact-exchange normal form rather than by a free taxonomy.

Definition 12.1 (Exact symmetric continuation claim). *An exact symmetric continuation claim for h on $\mathcal{S}$ asserts that, after irreversible loss, matter–antimatter exchangeability remains admissibly transportable as a same-history identity symmetry while the same scope continues to support persistent material reference.*

Lemma 12.2 (Exact-exchange continuation normal form). *Any post-irreversible exact-exchange continuation claim falls under the following decision tree.*

1. *Does the claimed exchange change standing-bearing carrier-role content? If no, it is skin exchange. If yes, continue.*
2. *Is there an exchangeability preservation certificate from the late admitted data of $\mathcal{S}$? If yes, the history has not undergone genuine retrieval-irreversibility for the exchange problem. If no, continue.*
3. *The claimed exchange must then be carried by hidden selector import, post-hoc reindexing, repaired exchange totalization, carrier-role motion, witness erasure, or scope change.*

Proof. The first branch is bivalent because an exchange either changes standing-bearing carrier-role content or it does not. If it does not, it changes no anchor quotient, role profile, continuation, retrieval status, or witness burden, and is skin. If it does, lawful same-scope exchange requires a certificate. If the certificate exists, genuine retrieval-irreversibility is absent by the certificate-equivalence lemma. If it does not, the claimed exchange must acquire force from something outside

the admitted same-scope data: selector, repair, reindexing, role motion, witness erasure, or scope change. These are exactly the normal forms enumerated in the route family. $\square$

Definition 12.3 (Symmetric-continuation route family). The following route labels expand the normal-form branches. A proposed exact symmetric continuation claim has a route image in the following finite family:

- R1. Latent global indexing:** the late history is claimed to retain hidden information sufficient to reconstruct the global exchange map.
- R2. Hidden selector restoration:** a selector, ranking, canonical representative, measure, or privileged pairing chooses a symmetry-preserving reindexing not fixed by the late anchor quotient.
- R3. Post-hoc global relabeling:** after irreversible loss, the labels “matter” and “antimatter” are globally swapped or reassigned to restore symmetry.
- R4. Oscillatory sign transport:** the sign side carrying the material-reference role is allowed to reverse or oscillate along the same admissible lineage.
- R5. Scope minimization:** the material-reference burden is weakened until no persistent anchor class is required.
- R6. Pure bookkeeping reinterpretation:** exact symmetry is retained only as a non-standing change of notation that changes no anchor quotient, witness set, or material-reference role.
- R7. Mechanism-level evasion:** a microphysical baryogenesis, antibaryogenesis, or symmetric-domain mechanism is invoked as though it by itself answered the fixed-domain transport question.

Theorem 12.4 (Nonempty route-image theorem). *Every exact symmetric continuation claim on a fixed scope with persistent material reference has nonempty route image in the route family **R1–R7**.*

Proof. The exact-exchange continuation normal form shows that every claim is either skin exchange, certified transport, or an uncertified standing-changing exchange. Skin exchange is **R6**. Certified transport negates genuine retrieval-irreversibility and appears as **R1** when stated as retained global indexing. Uncertified standing-changing exchange must be carried by hidden selector, post-hoc relabeling, oscillatory sign transport, scope minimization, or mechanism-level evasion, giving **R2–R5** or **R7**. Hence every exact symmetric continuation claim has a nonempty route image. $\square$

Theorem 12.5 (Route-collapse theorem). *No route in **R1–R7** survives as a same-scope exact symmetric continuation route.*

Proof. **R1** collapses by the certificate-equivalence lemma. If the global indexing remains recoverable on $\mathcal{S}$, genuine retrieval-irreversibility has not occurred. If it does not remain recoverable, the route imports hidden data.

R2 collapses because the selector is not fixed by the anchor quotient. If it affects standing-bearing material-reference content, it is a second same-scope classifier or hidden load. If it does not, it is skin and cannot restore exact standing-bearing symmetry.

R3 collapses by the no-global-reindexing lemma. Post-hoc relabeling after irreversible loss is not lawful same-history transport; it is a reset of the indexing relation.

R4 collapses by persistent-class conservation and the sign-fixation theorem. The sign side carrying the material-reference role cannot reverse on the same admissible lineage without illicit reindexing or scope change.

R5 exits the theorem. A scope with no persistent material-reference burden is not a same-scope counterexample to a theorem about chemistry-bearing or observer-bearing persistent material history.

R6 collapses to non-standing skin. If exact symmetry changes no anchor quotient, witness set, sign side, or downstream material-reference role, then it is notation only and carries no admissibility-bearing force.

R7 changes levels. Microphysical mechanisms can realize, suppress, dilute, or distribute projected particle-number asymmetry, but they do not by themselves restore lawful transport of a global exchange map through irreversible loss. If a mechanism supplies new standing-bearing data, the scope has been enriched. If it does not, it does not answer the transport problem. $\square$

Theorem 12.6 (Symmetric-continuation exhaustion theorem). *On any fixed SCMS scope requiring persistent material reference through irreversible evolution, exact matter–antimatter exchangeability has no surviving same-scope continuation route.*

Proof. By the exact-exchange continuation normal-form lemma and the nonempty route-image theorem, any exact symmetric continuation claim has at least one route in **R1–R7**. By the route-collapse theorem, no route survives as a same-scope exact symmetric continuation route. Therefore exact matter–antimatter exchangeability has no surviving same-scope continuation route under the stated scope conditions. $\square$

Corollary 12.7 (No semantic-relabelling escape). *A claim that “matter” and “antimatter” are merely names does not threaten the theorem. If the renaming changes no standing-bearing anchor content, it is skin. If it changes which late material-reference classes stand, it is post-hoc reindexing or hidden selector import and is inadmissible on the fixed scope.*

Proof. A mere name change is route **R6**; it is non-standing. A name change that determines late material-reference roles is route **R2** or **R3**; those routes collapse by the route-collapse theorem. $\square$

13 Surplus-Class Redescription and Amplification Exhaustion

The symmetric-continuation routes concern exact exchangeability. A different escape route claims that, after $\mathfrak{S}_S^{\text{uns}}(h)$ is fixed at $\mathcal{B}_S(h)$, same-scope redescription, feedback, or repeated reconstruction can increase or reauthorize the standing-bearing surplus reconstruction class. That is not an exchangeability-preservation route; it is a surplus-class redescription route.

Definition 13.1 (Surplus-class amplification claim). *A surplus-class amplification claim asserts that, after $\mathfrak{S}_S^{\text{uns}}(h)$ is fixed at $\mathcal{B}_S(h)$, same-scope admissible redescription, bookkeeping, feedback, or repeated reconstruction can increase, duplicate, erase, or reauthorize the standing-bearing surplus reconstruction class without declaring new tensor load, source dynamics, boundary data, or scope change.*

Lemma 13.2 (Surplus-class redescription normal form). *Any alleged post-boundary amplification of $\mathfrak{S}_S^{\text{uns}}(h)$ falls under the following decision tree.*

1. *Does the alleged amplification change the defining tuple*

$$(\mathcal{Q}_S(h), \mathcal{Q}_S^{\text{bur}}(h), \text{Role}_h^S, \text{Cont}_h, \text{Rec}_\Xi(h), \mathcal{W}_S(h), \text{arank}_S(h)) / \sim_{\text{adm}},$$

with $\Delta_B^S(h)$ added only for $\mathfrak{S}_S^{\text{SCMS}}(h)$?

If no, it is bookkeeping. If yes, continue.

2. *Is the change declared as physical dynamics, tensor load, boundary data, source exchange, or scope enrichment? If yes, it is mechanism-level physical realization or richer-scope dynamics, not same-scope redescription amplification. If no, it is hidden load, selector import, repaired totalization, or standing failure.*

Proof. Downstream admissible use may inherit, condition on, transform, or enrich a fixation, but it cannot erase or reauthorize the fixation as the same act. Therefore an alleged amplification either preserves the defining tuple of $\mathfrak{S}_S^{\text{uns}}(h)$, in which case it is bookkeeping, or changes that tuple. The SCMS signed specialization also preserves $\Delta_B^S(h)$ after sign fixation. A changed tuple must be declared as physical dynamics, tensor load, boundary data, source exchange, or scope enrichment; otherwise it is hidden load, selector import, repaired totalization, or standing failure. The route labels are expanded instances of this normal form. $\square$

Definition 13.3 (Surplus-class redescription route family). The following route labels expand the normal-form branches. A surplus-class amplification claim has a route image in the following family:

Route	Claim	Collapse
A1	Redescription amplification	Admissible redescription increases the surplus reconstruction class. It is bookkeeping if witnesses are unchanged and scope/tensor change if witnesses are changed.
A2	Feedback accumulation	Repeated same-scope feedback accumulates new standing surplus. It is either declared physical dynamics or hidden tensor load.
A3	Mechanism-label amplification	Naming a process “baryogenesis” supplies standing. The label is skin unless the mechanism supplies declared witnesses.
A4	Entropy/probability amplification	Entropy, typicality, or probability generates the class. Those are downstream witnesses or measures, not admissibility sources.
A5	Conservation-law laundering	Physical conservation language is used to prove redescription invariance. This is a category error unless stated as structural redescription invariance.
A6	Numerical abundance laundering	A value of η_B is inferred from structural class fixation. This is unsupported without dynamical witnesses.

Theorem 13.4 (Surplus-class amplification exhaustion). *No same-scope admissible redescription route increases, duplicates, erases, or reauthorizes the boundary-fixed surplus reconstruction class. If the route changes no anchor quotient, role profile, continuation relation, retrieval status, or witness family, it is bookkeeping. If it changes any of them, it is declared physical dynamics, tensor load, anchor motion, repaired totalization, or scope change.*

Proof. The surplus-class redescription normal form exhausts the possibilities. A same-scope admissible redescription preserves the data defining $\mathfrak{S}_S^{\text{uns}}(h)$: anchor quotient, burden-realizing support, role profile, continuation relation, retrieval status, witness family, and anchor rank. If an alleged amplification preserves all of those data, it adds no standing-bearing content and is bookkeeping. If it changes one of those data, it has changed tensor content, moved an anchor, repaired an undefined exchange operation, imported a selector, or changed scope. Entropy, probability, mechanism labels, and numerical abundance claims can participate only through declared physical ledgers or parameter modules; they do not generate admissibility of the class by redescription. $\square$

Corollary 13.5 (Surplus-class invariance under admissible redescription). *Let $\mathfrak{S}_S^{\text{uns}}(h)$ be the boundary-fixed unsigned admissible surplus reconstruction class, and let $\mathfrak{S}_S^{\text{SCMS}}(h)$ denote its signed SCMS specialization when SCMS connectivity is declared. Any same-scope admissible redescription preserves the relevant class up to lawful equivalence. It cannot increase, duplicate, erase, or reauthorize the class. Apparent growth, depletion, transfer, dilution, or conversion of projected particle-number asymmetry belongs to declared physical dynamics inside a mechanism-level realization; it is not an increase in the admissibility of the reconstruction class.*

Proof. The corollary is the invariant form of the amplification exhaustion theorem. Same-scope redescription preserves the class. Change without new declared physical or reconstruction load is bookkeeping; change with new load is tensor, anchor, repair, or scope motion. Therefore admissible redescription cannot amplify the surplus reconstruction class. $\square$

This is a redescription-invariance result, not a physical conservation law for baryon number or particle abundance.

Theorem 13.6 (Boundary-fixation theorem: general form). *Let $\mathcal{S}$ be a stable coherent material-reference scope and let h be a history undergoing genuine retrieval-irreversibility while realizing the declared material-reference burden. Then the unsigned admissible surplus reconstruction class*

$$\mathfrak{S}_S^{\text{uns}}(h)$$

is fixed at the persistent-material boundary regime $\mathcal{B}_S(h)$.

Proof. The pre-witness exchange-totalization obstruction proves that exact exchange cannot be totalized on the burden-realizing post-irreversible support without repair, selector import, or scope change. Weak obstruction-to-witness extraction gives witness nonemptiness when $\mathcal{O}_{\text{ex}}^{\text{bur}} \neq 0$ on a nonempty burden support. Burden-wide lower-bound extraction supplies the ρ_S -bound when $\mathcal{O}_{\text{ex}}^{\text{vbur}}$ holds. Symmetric-continuation exhaustion closes routes that would restore exact exchangeability, and surplus-class amplification exhaustion closes routes that would increase, erase, or reauthorize the unsigned class by redescription. The only same-scope admissible status left is that $\mathfrak{S}_S^{\text{uns}}(h)$ is fixed at the dependency-minimal package $\mathcal{B}_S(h)$. $\square$

Corollary 13.7 (Boundary-fixation theorem: SCMS signed specialization). *If $\mathcal{S}$ is an SCMS scope, then the signed specialization*

$$\mathfrak{S}_S^{\text{SCMS}}(h)$$

is fixed at $\mathcal{B}_S(h)$ after the sign-fixation theorem applies.

Proof. The general theorem fixes the unsigned reconstruction class. On an SCMS scope, sign fixation identifies the sign side carrying the connected material-reference burden and prevents later sign reversal without reindexing or scope change. Therefore the signed coordinate $\Delta_B^S(h)$ is fixed as the SCMS specialization of the boundary-fixed class. $\square$

14 Sakharov Conditions, CPT, Entropy, and Mechanism Status

14.1 Bridge to standard baryogenesis programs

The structural proof is complete before any mechanism-specific CP, sphaleron, reheating, freeze-out, or leptogenesis content is invoked. These structures are not premises of exchangeability non-transportability. They are possible physical realization, transfer, protection, dilution, or conversion routes for projected particle-number asymmetry inside an already boundary-fixed admissible surplus reconstruction class.

Concrete baryogenesis mechanisms such as freeze-out, electroweak baryogenesis, leptogenesis, sphaleron conversion, inflationary dilution, reheating, and entropy-producing dispersal may generate, transfer, protect, dilute, or convert projected particle-number asymmetry in a declared physical model. Their AASC status is that they occupy and witness the admissible surplus reconstruction class already forced at the persistent-material boundary regime; they do not generate the admissibility of that class by later redescription. The mechanism literature is cited here only to locate the physical-realization interface, not to review baryogenesis in full or to make any mechanism-specific numerical claim [6, 7, 8, 9, 10].

- **Freeze-out and departure from equilibrium.** Freeze-out supplies a familiar dynamical setting in which reaction networks stop erasing asymmetry. In the present terminology, it is a mechanism-level way of making projected particle-number asymmetry persistent enough to witness the late anchor quotient.
- **Electroweak baryogenesis.** Electroweak baryogenesis attempts to realize projected asymmetry through electroweak-scale nonequilibrium dynamics, sphaleron processes, and additional CP-odd mechanism content. The AASC theorem does not supply those rates; it states the structural burden such rates must satisfy.
- **Leptogenesis.** Leptogenesis realizes the burden indirectly: a lepton asymmetry is generated first and then partially converted into baryon asymmetry by Standard-Model sphaleron dynamics. In the present language, the persistent baryonic anchor class is reached through a transport chain rather than by immediate baryon-side production.
- **Inflationary dilution and reheating.** Inflation can dilute pre-existing relics and reheating can repopulate the relevant sectors. These processes affect which candidate histories enter the admissible post-irreversible anchor analysis; they do not remove the need for a late persistent anchor witness, sign-fixed on SCMS scopes, if persistent material reference is to survive. Reheating, where relevant, should be typed as a role-transition certificate into particle, gauge, mass/inertia, and record-bearing structures, not as primitive conversion magic.
- **Entropy production.** Entropy production is not used as a primitive metaphysical arrow. It is a physical witness for the loss of admissibly recoverable exchange-indexing correlations on the declared scope.

Proposition 14.1 (Cosmology bridge principle). *A concrete baryogenesis model is AASC-compatible on the present target precisely when it supplies dynamical witnesses for at least four items: projected asymmetry production or transfer, survival through washout, sign-side persistence into the late anchor quotient, and compatibility with the declared scope’s recoverability limits.*

Proof. Projected production or transfer is needed because the no exact symmetric persistent continuation theorem excludes exact exchangeable persistence. Survival through washout is needed because a transient asymmetry that leaves no persistent anchor witness does not realize the late material-reference burden. Sign-side persistence is needed because sign fixation is what turns a surplus witness into a stable carrier-role lineage. Compatibility with recoverability limits is needed because a model that restores exact exchange transport after irreversible loss either has not instantiated the theorem’s irreversibility premise or has changed scope. These are structural requirements on the model’s role; the microphysical details remain the province of standard cosmology. $\square$

Proposition 14.2 (Structural necessity versus dynamical realizability). *The present argument does not replace the Sakharov conditions. It structurally reinterprets their role. The structural necessity of a nonzero admissible surplus reconstruction class is upstream of mechanism choice; the Sakharov conditions remain dynamical side-conditions on how projected asymmetry can be realized in conventional cosmological model-building.*

Proof. The structural theorem proves that exact exchangeability is inadmissible under irreversible history once persistent material reference is required. It does not specify which microphysical process generates projected asymmetry. Baryon-number violation, C/CP violation, and nonequilibrium dynamics remain conditions on mechanism realizability inside standard dynamical frameworks. They are therefore downstream of the structural necessity rather than identical with it. $\square$

Proposition 14.3 (Mechanism occupancy of the forced class). *A concrete baryogenesis mechanism is AASC-compatible on the present target precisely when it supplies declared physical witnesses for projected production or transfer, washout survival, sign-side persistence, late material-reference transport, and compatibility with the recoverability limits of the declared scope. Such a mechanism may generate particle-number asymmetry in its own physical model; it does not generate the admissible surplus reconstruction class’s admissibility from outside the boundary-fixed reconstruction class.*

Proof. If a mechanism supplies production, conversion, washout control, sign persistence, and late transport witnesses, then it realizes the physical side of the already-fixed class. If it lacks those witnesses, it does not realize the persistent material-reference target. If it claims to generate the class’s admissibility by later redescription, it instantiates a surplus-class amplification route or scope change. Therefore mechanisms may be physically generative at the projected particle-number level while remaining occupants and witnesses of the forced admissible class at the reconstruction level. $\square$

14.2 Mass/inertia as projection carrier of the surplus reconstruction class

The boundary-fixed unsigned surplus reconstruction class $\mathfrak{S}_{\text{g}}^{\text{uns}}(h)$ is not identical to a particle species, mass value, stress-energy tensor, or Standard Model parameter. When projected into a mature realized physical interior, it is represented through localized particle persistence classes and their deformation-response / transport-facing profiles. This is a projection into physical realization, not a derivation of numerical masses or a privilege assigned to any particular quark sector.

The projection from $\mathfrak{S}_S^{\text{uns}}(h)$ to $\Pi_D \rightarrow \mu_D \rightarrow \iota_D$ is a realization map from the reconstruction class into the mature physical interior; it is not an identity map and not a numerical mass derivation. In realized-physical projection, the carrier chain is:

$$\mathfrak{S}_S^{\text{uns}}(h) \longrightarrow \Pi_D \longrightarrow \mu_D(\Pi) \longrightarrow \iota_D(\Pi) \longrightarrow \text{Ledgers}_D^{\text{source/interaction/record}}.$$

Here Π_D is a localized particle persistence class, $\mu_D(\Pi)$ is its deformation-response profile, and $\iota_D(\Pi)$ is its transport-facing restriction.

Proposition 14.4 (Mass/inertia projection firewall). *Mass and inertia may carry projected realizations of the admissible surplus reconstruction class through deformation-response and transport-facing profiles, but a scalar mass value is not by itself a source of that class's admissibility.*

Proof. A scalar mass value becomes source- or carrier-bearing only when it represents a licensed deformation-response profile of a localized persistence class inside a field/gauge/transport ledger. Without that role structure, it is an untyped parameter or display value. With that role structure, it can participate in physical realization of $\mathfrak{S}_S^{\text{uns}}(h)$, but it does not generate that class's admissibility. Therefore mass/inertia carries and witnesses projected realizations only through declared deformation-response and transport-facing ledgers. $\square$

The present theorem does not single out any particular quark sector, any Yukawa value, any CKM coordinate, any CP phase value, or any sector-specific mass sensitivity. Quark-sector masses, Yukawa structures, and mixing/phase slots are mechanism-facing projection content unless a separate theorem supplies stronger support.

14.3 CP-odd structure as mechanism-facing content

CP-odd structure is not part of the proof of the boundary-fixed surplus class. It enters only at the mechanism-interface layer. The structural theorem requires no numerical CP phase, no CKM coordinate, no PMNS coordinate, no strong-CP resolution, and no CP-sector numerical derivation. CP-odd content enters only when a concrete physical baryogenesis mechanism claims to realize, transfer, protect, or convert projected asymmetry inside a declared model.

Definition 14.5 (Admissible CP-odd mechanism content). CP-odd content is admissible in a mechanism-level realization only when it is carried by a declared invariant, quotient-stable phase relation, holonomy-like obstruction, mixing/phase slot, or standing-bearing tensor content inside the mechanism scope. A free phase label that changes no standing-bearing witness is skin; a free phase label that changes physical content without a declared ledger is hidden tensor load.

Proposition 14.6 (CP parameter-status firewall). *CP-odd quantities may be necessary for concrete baryogenesis dynamics, but they are not primitive sources of the admissible surplus reconstruction class. Their numerical values remain downstream parameter targets unless the scope supplies a witness, exactness certificate, quotient collapse, reconstruction rule, or zero-anchor derivation.*

Proof. A CP-odd quantity contributes to a mechanism only if it affects standing-bearing transition, interaction, mixing, or witness content in the declared model. If it affects no witness, it is skin. If it affects witnesses without a declared ledger, it is hidden tensor load. If it is declared, it is part of the mechanism-level tensor or parameter content. In none of these cases does the CP-odd value generate the admissibility of the surplus reconstruction class from outside the boundary regime. $\square$

Proposition 14.7 (CPT firewall). *The theorem does not assert violation of CPT symmetry at the level of microscopic interactions. A CPT-invariant dynamical theory may still realize a persistent anchor witness, sign-fixed on SCMS scopes, once the cosmological history and its admissible material-reference scope are fixed.*

Proof. CPT invariance is a statement about the symmetry structure of the fundamental interaction laws under charge conjugation, parity inversion, and time reversal. The present theorem concerns a different object: the admissible persistence of late-time material-reference anchors under irreversible loss of retrievable correlation. Structural sign fixation in the late anchor quotient does not imply that the microscopic equations fail CPT invariance. It implies only that the exact exchange symmetry is not lawfully transportable as the same-history material-reference symmetry after irreversible loss. $\square$

Proposition 14.8 (Entropic dispersal as witness-destroying transport failure). *Thermodynamic entropy increase is relevant to the present theorem only insofar as it supplies a physical witness for retrieval-irreversibility: dispersal of correlations can destroy the admissible transportability of a global exchange map.*

Proof. The theorem does not require a complete thermodynamic derivation of the arrow of time. It requires that the fixed scope fail to contain the correlations needed for lawful global relabeling. Entropic dispersal is a standard physical route by which detailed correlations become unrecoverable for late-time material-reference practice. When that happens, it witnesses the precise admissibility failure used by the no-global-reindexing lemma. If a model instead preserves a lawful recovery map, then it has not instantiated the irreversibility premise on that scope. $\square$

Proposition 14.9 (Domain and anti-domain histories). *A cosmology containing disconnected matter-rich and antimatter-rich domains is not automatically a counterexample. Either the domains are separate indexed material-reference scopes, or a transport certificate must show that exact global exchangeability remains lawful through irreversible loss while preserving one fixed late-time anchor quotient.*

Proof. If domains do not participate in one connected SCMS burden, they are not a same-scope counterexample to the SCMS theorem. If they are claimed to form one fixed scope, then the same anchor quotient, SCMS connectivity, and no-global-reindexing constraints apply. A domain model therefore either changes the scope indexing or must supply the exact transport certificate excluded by the symmetric-continuation exhaustion theorem. $\square$

15 Objections and Clarifications

The following objections identify the main interpretive and proof-register pressure points. Each response states the theorem-level distinction that blocks the objection.

“You imposed the AASC kernel on cosmology.”

The kernel-instantiation theorem states the opposite: any domain already requiring stable coherent material reference automatically instantiates admissibility, standing, reference, and irreversibility.

“This is just semantics; labels can be swapped.”

A swap that changes no anchor content is skin. A swap that changes standing-bearing material reference is post-hoc reindexing or selector import.

“This is tautological.”

The nontrivial claim is not that matter exists, but that exact global exchangeability cannot remain a standing-preserving transport invariant after retrieval-irreversibility without a certificate.

“Microscopic dynamics may be reversible.”

The theorem requires retrieval-irreversibility on the declared scope, not fundamental dynamical irreversibility. Exact external inversion is not a same-scope recovery map.

“This is anthropic.”

The theorem is conditional on stable coherent material reference, not observer selection. Chemistry-bearing persistence already instantiates the relevant burden.

“CPT symmetry forbids the result.”

The result is not microscopic CPT violation. It is late-time anchor-sign fixation under irreversible loss of transportability.

“The Sakharov conditions already explain baryogenesis.”

They remain mechanism-realizability conditions. The present result states why an admissible surplus reconstruction class, represented by a persistent anchor witness, is structurally required on the burden-realizing support when exchange transport fails there; the scope-indexed lower bound follows only under burden-wide exchange non-transportability, and sign fixation follows on SCMS scopes.

“The boundary class is introduced before its data exist.”

The formal definition is placed after the anchor quotient and $C_{\text{BAR}}^{\text{pre}}/C_{\text{BAR}}^{\text{wit}}/Q_{\text{BAR}}$ construction. Earlier boundary language is only dependency orientation.

“The first boundary is undefined.”

The boundary regime $\mathcal{B}_S(h)$ is minimal under the reconstruction-status dependency preorder $\preceq_{\text{dep}}$, not a temporal first instant.

“CBAR bakes in the surplus witness.”

The nonempty persistent anchor witness is extracted from pre-witness obstruction plus anchor realization before witness-preserving equivalence is imposed. The witness-preserving category $C_{\text{BAR}}^{\text{wit}}$ is a post-extraction audit quotient.

“Boundary fixation uses later theorems.”

The early boundary statement is conditional; the boundary-fixation theorem appears after exchange obstruction, sign fixation, symmetric-continuation exhaustion, and surplus-class amplification exhaustion.

“Retrieval-irreversibility is vague.”

For the exchange problem, genuine retrieval-irreversibility is equivalent to absence of an exchangeability preservation certificate from late admitted data.

“Route tables are arbitrary.”

The route tables are generated by binary normal-form decision trees: standing-changing or skin; certificate or no certificate; declared dynamics or hidden load.

“Nonzero means abundance.”

By convention, nonzero means nonempty anchor witness or nonvanishing obstruction unless particle abundance is explicitly named.

“Amplification is not an exchangeability route.”

Correct. Amplification is handled in a separate surplus-class redescription exhaustion theorem, not in symmetric-continuation exhaustion.

“You do not derive the baryon-to-photon ratio.”

Correct. The paper derives structural existence, sign fixation, and a scope-indexed lower bound. Numerical abundance requires additional dynamics.

“You are denying baryogenesis mechanisms.”

No. Mechanisms may generate projected particle-number asymmetry; they do not generate the admissibility of the boundary-fixed surplus reconstruction class.

“Mass/inertia language is appended.”

It enters only at the physical-realization projection layer: $\mathfrak{S}_S^{\text{uns}}(h) \rightarrow \Pi_D \rightarrow \mu_D \rightarrow \iota_D$, not in the exchangeability proof itself.

“CP violation is doing hidden proof work.”

No. CP-odd content is mechanism-facing only; the structural theorem requires no CP numerical value.

“This is a conservation law in disguise.”

No. Surplus-class invariance is redescription invariance, not physical baryon-number conservation.

“An antimatter universe would also be possible.”

Yes. The theorem fixes a sign, not a metaphysical preference for the sign named matter. Naming the surviving side “matter” is convention after sign fixation.

“Matter and antimatter domains might coexist.”

Either they are separate indexed scopes or the model owes a transport certificate for one fixed SCMS scope.

“A hidden exchange sector carries the real symmetry.”

If it preserves all prior witnesses, conservative reconstruction collapse makes it bookkeeping. If it changes a witness, it is selector import, repaired totalization, carrier laundering, or scope transport.

“The universal quotient is formal machinery.”

The quotient Q_{BAR} states the exact same-scope identification: every admissible reconstruction preserving the prior witness package factors through it uniquely. It adds no dynamics or numerical parameters.

Additional closure responses. Residual attacks are closed by the exchangeability, boundary-fixation, and redescription-invariance layers. First, global unitarity or total-state retention is insufficient: global retention is not same-scope admissible recoverability unless the model supplies an exchangeability preservation certificate from the late scope’s admitted data. Second, symmetric persistent structures are admissible only in the skin case or certified-transport case; after genuine retrieval-irreversibility, neither yields standing-bearing exact symmetric continuation. Third, pure sign-label renaming is skin, whereas carrier-role motion requires lawful transport and otherwise collapses. Fourth, standard baryogenesis mechanisms remain important, but they supply dynamical witnesses for projected production, survival, sign persistence, and washout control; they do not replace the fixed-domain surplus-class burden. Fifth, post-boundary amplification is treated separately from exact exchangeability survival. Sixth, mass/inertia and CP-odd structures are projection-facing mechanism interfaces rather than proof-spine premises.

16 What the Paper Does Not Claim

This paper does *not* claim:

- a derivation of the observed baryon-to-photon ratio,
- a selection of one baryogenesis mechanism over another,
- a violation of CPT symmetry at the level of fundamental interactions,
- a replacement for Standard Model or beyond-Standard-Model baryogenesis dynamics,
- a proof that all imaginable formal universes must contain matter dominance,
- a denial that an antimatter-dominated sign convention could have been fixed instead,
- a denial that exact total-state descriptions may contain correlations unavailable to same-scope material-reference recovery,
- a derivation of CKM, PMNS, Jarlskog, strong-CP, or other numerical CP-sector values,
- a theorem singling out any specific quark sector as the privileged carrier of the surplus reconstruction class,
- a physical conservation law for baryon number or particle number beyond declared mechanism dynamics,
- a claim that the boundary regime is a first temporal instant or physical origin,
- a claim that physical particle-number surplus is numerically fixed at the boundary regime.

Those non-claims are load-bearing. The paper is a fixed-domain structural theorem about admissible cosmic history, not a mechanism-selection paper, not a value-derivation paper, and not a CPT-violation proposal.

Its claim is narrower and, within that scope, stronger: a persistent anchor witness, carried by the admissible surplus reconstruction class, is forced on the burden support when exact exchange transport fails there; the scope-indexed lower bound follows under burden-wide exchange non-transportability.

17 Conclusion

The theorem does not derive the numerical baryon asymmetry, select a baryogenesis mechanism, determine CP phase values, or privilege a particle sector. It proves a fixed-domain reconstruction result: in a stable coherent material-reference scope undergoing genuine retrieval-irreversibility, exact matter–antimatter exchangeability is not transportable as a late same-history identity symmetry on the burden-realizing support. Nonvanishing of the burden-restricted obstruction gives witness nonemptiness; burden-wide exchange non-transportability gives the scope-indexed lower bound. The unsigned admissible surplus reconstruction class is therefore fixed at the persistent-material boundary regime; on SCMS scopes its signed specialization is fixed as well. Concrete physical mechanisms may generate, transfer, protect, dilute, or convert projected particle-number asymmetry inside that class; they do not generate the admissibility of the class by later redescription.

The surviving asymmetry claim is consequently structural and quotient-level. It says that the late material-reference lineage cannot be carried by exact global exchange symmetry once the exchange certificate is unavailable on the declared scope. It does not say that the sign named “matter” is metaphysically preferred, nor that a particular physical mechanism is selected. It says that persistent material identity requires a burden-supported persistent anchor witness, with the lower-bound rank supplied by burden-wide exchange non-transportability, whose sign is fixed on SCMS scopes and whose witness-stage reconstruction is unique up to admissible bookkeeping equivalence. Later physical mechanisms occupy and witness that boundary-fixed class.

18 Baryogenesis Reconstruction Degeneracy Matrix

ID	Degeneracy / escape route	Blocking result
D1	Exact matter–antimatter exchange transported through the post-irreversible anchor quotient	Exchange totalization obstruction; nonvanishing of $\mathcal{O}_{\text{ex}}$
D2	Hidden global index reconstructs the early exchange pairs	Retrieval-status preservation; no-global-reindexing; B1/B2 obstruction
D3	Matter/antimatter labels are swapped after late-time structure forms	Carrier-role versus sign-label distinction; post-hoc reindexing route collapse
D4	Symmetric domains are treated as one SCMS scope	Domain and anti-domain scope theorem; requires transport certificate or scope indexing
D5	A microphysical mechanism is invoked as a complete answer to the transport problem	Mechanism-level bridge; mechanisms supply witnesses but do not replace same-scope reconstruction
D6	A hidden symmetry witness preserves exact exchangeability while all prior witnesses remain fixed	Conservative reconstruction collapse for hidden exchange sectors
D7	Exact symmetry is kept as pure notation	Skin covariance; bookkeeping duplication B6
D8	Sign side reverses along the same material lineage	Persistent-class conservation and sign-fixation theorem

ID	Degeneracy / escape route	Blocking result
D9	Late re-symmetrization restores exact exchangeability	No late re-symmetrization theorem; no exact symmetric persistent continuation theorem
D10	Numerical baryon abundance is demanded from the theorem	Scope-indexed lower bound is structural, not phenomenological
D11	Global unitarity or exact microstate retention is treated as recovery	Admissible recoverability certificate requirement; global retention versus same-scope recovery distinction
D12	CPT invariance is claimed to forbid sign fixation	CPT firewall: structural anchor-sign fixation is not microscopic CPT violation
D13	Post-boundary feedback or redescription is claimed to amplify the standing-bearing surplus reconstruction class	Surplus-class amplification exhaustion and surplus-class invariance under admissible redescription; declared physical dynamics may change projected particle-number content but not class admissibility

19 Symbol Ledger

Symbol	Meaning
$\mathcal{S}$	Fixed cosmological material-reference scope.
$\mathcal{H}$	Class of candidate histories in the declared scope.
h	A candidate cosmological history.
$t_*(h)$	Stage after which exchange-relevant correlations are no longer admissibly recoverable on $\mathcal{S}$.
$\mathcal{R}$	Standing-bearing material-reference classes required by the scope.
$\mathcal{C}_{\mathcal{S}}(h)$	Persistent anchor candidates after the irreversible stage.
$\mathcal{Q}_{\mathcal{S}}(h)$	Persistent anchor quotient by sign type and downstream role profile.
$\text{Role}_h^{\mathcal{S}}$	Downstream material-reference role map.
$\mathcal{W}_{\mathcal{S}}^{\text{M}}, \mathcal{W}_{\mathcal{S}}^{\bar{\text{M}}}$	Matter-side and antimatter-side persistent anchor witness sets; notation is fixed before nonemptiness is proved.
$\text{arank}_{\mathcal{S}}(h)$	Total number of persistent anchor witness classes.
$\Delta_B^{\mathcal{S}}(h)$	Signed persistent anchor witness.
$\rho_{\mathcal{S}}$	Minimum persistent anchor burden imposed by the declared scope.
$\mathcal{C}_{\text{BAR}}^{\text{pre}}$	Pre-witness category of finite persistent-carrier baryogenesis reconstructions.
$\mathcal{C}_{\text{BAR}}^{\text{wit}}$	Witness-preserving audit category after the persistent anchor witness is fixed.
$\mathcal{C}_{\text{BAR}}^{\partial}$	Partial reconstruction category with explicit exchange-operator domains.
$\mathcal{Q}_{\text{BAR}}^{\text{pre}}, \mathcal{Q}_{\text{BAR}}, \mathcal{Q}_{\text{BAR}}^{\partial}$	Pre-witness, witness-preserving, and partial-domain reconstruction quotients.
X_h	Partial matter–antimatter exchange operator.
$\mathcal{O}_{\text{ex}}$	Internal exchangeability totalization obstruction.
$\text{Env}_{\mathcal{S}}$	Envelope data not derived by the theorem: mechanisms, rates, thermal history, and numerical inputs.
$\mathcal{B}_{\mathcal{S}}(h)$	Persistent-material boundary regime: dependency-minimal licensed reconstruction status for the material-reference surplus package.
$\mathfrak{S}_{\mathcal{S}}^{\text{uns}}(h)$	Boundary-fixed unsigned admissible surplus reconstruction class, not a numerical abundance.
$\mathfrak{S}_{\mathcal{S}}^{\text{SCMS}}(h)$	Signed SCMS specialization of the boundary-fixed reconstruction class.
$\text{TC}_{\mathcal{S}}^{\Delta}$	Surplus transport/closure relation from the boundary-fixed surplus reconstruction class to late material-reference witnesses.
Π_D, μ_D, ι_D	Particle persistence class, deformation-response profile, and transport-facing inertia restriction used in realization.

20 Proof Dependency Graph

- G1. Stable coherent material reference instantiates the AASC kernel.
- G2. Fixed cosmological scope and exchange-relevant retrieval-irreversibility are defined.
- G3. “Nonzero” is fixed as anchor-witness nonemptiness or obstruction nonvanishing, not numerical abundance.
- G4. Anchor equivalence defines the canonical anchor quotient.
- G5. $\mathcal{C}_{\text{BAR}}^{\text{pre}}$ locks the pre-witness reconstruction comparison class without preserving $\Delta_B^{\mathcal{S}}(h)$ as an input.
- G6. $\mathcal{C}_{\text{BAR}}^{\partial}$ makes exact exchangeability a partial operator with a fixed admissible domain.

- G7. Exchange totalization obstruction blocks global late exchange on the declared carrier; its burden-restricted form acts on $\mathcal{Q}_S^{\text{bur}}(h)$.
- G8. Genuine retrieval-irreversibility is equivalent, for this exchange problem, to absence of an exchangeability preservation certificate from late admitted data.
- G9. Nonvanishing of $\mathcal{O}_{\text{ex}}$ blocks full-carrier totalization; nonvanishing of $\mathcal{O}_{\text{ex}}^{\text{bur}}$ marks some totalization failure on the burden-realizing support; $\mathcal{O}_{\text{ex}}^{\text{vbur}}$ records burden-wide failure on every minimum burden-realizing subfamily.
- G10. Weak obstruction-to-witness extraction combines $\mathcal{O}_{\text{ex}}^{\text{bur}} \neq 0$ with nonempty burden support to obtain witness nonemptiness.
- G11. $\mathcal{C}_{\text{BAR}}^{\text{wit}}$ and $\mathcal{Q}_{\text{BAR}}$ audit witness-preserving equivalence after the witness has been extracted.
- G12. The reconstruction-status preorder $\preceq_{\text{dep}}$ defines the boundary regime $\mathcal{B}_S(h)$.
- G13. The unsigned boundary-fixed admissible surplus reconstruction class $\mathfrak{S}_S^{\text{uns}}(h)$ is defined without fixing numerical abundance; $\mathfrak{S}_S^{\text{SCMS}}(h)$ adds the signed coordinate on SCMS scopes.
- G14. Anchor realization plus $\mathcal{O}_{\text{ex}}^{\text{vbur}}$ gives the burden-wide lower-bound theorem $\text{arank}_S(h) \geq \rho_S$.
- G15. On SCMS scopes, sign fixation yields $|\Delta_B^S(h)| \geq \rho_S$.
- G16. The exact-exchange normal-form lemma generates the symmetric-continuation route family.
- G17. Symmetric-continuation exhaustion states final non-survival of exact exchangeability.
- G18. The surplus-class redescription normal-form lemma generates the amplification route family.
- G19. Surplus-class amplification exhaustion blocks post-boundary growth of the admissible class while allowing declared physical particle-number dynamics.
- G20. The boundary-fixation theorem fixes $\mathfrak{S}_S^{\text{uns}}(h)$ at $\mathcal{B}_S(h)$ and fixes $\mathfrak{S}_S^{\text{SCMS}}(h)$ there on SCMS scopes.
- G21. Surplus transport/closure carries the boundary-fixed class to late material-reference witnesses.
- G22. Mass/inertia and admissible CP-odd mechanism content classify how physical mechanisms may carry and witness projected realizations of the forced class.

21 AASC Source-Use Summary

The following AASC closure roles are used here.

- *Kernel necessity*: stable coherent reference automatically instantiates admissibility, standing, reference, and irreversibility.
- *AMetric boundary*: no selector, ranking, canonical representative, probability, typicality rule, or hidden index has boundary authority unless fixed by the declared domain.

- *Anchor/tensor/skin*: material-reference burdens are anchors; role profiles, anchor quotients, continuation, and surplus witnesses are tensor content; sign names and representatives are skin when witness-inert.
- *Fixed-domain exhaustion*: same-scope variation must change anchors, tensor witnesses, quotient identity, continuation, or scope. No additional governance-bearing locus remains.
- *Partial-domain closure*: an undefined same-scope operator application cannot be made defined on the same carrier by admissible extension; doing so is repair/generation or scope change.
- *Conservative reconstruction collapse*: hidden symbols preserving all prior witnesses collapse to definitional bookkeeping unless they import new standing-bearing load.
- *No-repair irreversibility*: lost exchange transport cannot be restored later as the same admissible history by post-hoc relabeling.
- *Boundary discipline*: the persistent-material boundary regime is a dependency-minimal licensing status, not a first temporal instant or primitive cause.
- *Surplus-class invariance*: admissible redescription preserves the boundary-fixed surplus reconstruction class; it does not physically conserve baryon number or particle abundance.
- *Transport/closure*: late coherent witnesses require lineage from the boundary-fixed surplus reconstruction class when independent local continuation is insufficient.
- *Mass/inertia role*: localized particle persistence classes carry deformation-response and transport-facing profiles rather than primitive scalar mass sourcehood.
- *CP and parameter status*: CP-odd mechanism content is mechanism-facing and must enter as declared invariant, quotient-stable, or parameter-level tensor content, not as a free primitive source.

22 Source-Dependency Ledger

This ledger records the source roles used for boundary discipline, transport/closure, mass/inertia projection, CP-odd parameter discipline, and redescription invariance. Each imported source is used only at the level stated here.

Early metric-interior transport closure [11].

Supplies boundary-regime discipline and the transport/closure certificate analogy. It does not turn the baryogenesis theorem into an inflation theorem, and it does not assert a scalar-field mechanism, e-fold count, or reheating value.

Arrow of time / standing monotonicity [12].

Supplies no-backflow, standing-monotone retention, and the rule that downstream use cannot erase upstream fixation as the same act. It does not derive entropy production, CP values, or baryogenesis rates.

Realized physical interior [13].

Supplies particle persistence, fields, gauge, mass/inertia, vacuum, time, causality, and record roles as the physical role ladder. It does not derive numerical masses, CKM/PMNS values, or a full parameter table.

Einstein/source-ledger projection [14].

Supplies the source-side chain from particle persistence through deformation-response and inertia into metric-facing ledgers. It does not make stress-energy, scalar mass, or vacuum energy primitive.

Target $C_{\text{BAR}}^{\text{pre}}/C_{\text{BAR}}^{\text{wit}}/Q_{\text{BAR}}$ machinery.

Supplies the local same-scope quotient, burden-restricted obstruction, route exhaustion, SCMS sign fixation, and lower-bound burden. It does not derive the baryon-to-photon ratio or select a mechanism.

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